

DEFINITION BREAKER

A puzzle-adventure and learning system
where the rules are physical objects

Touch. Test. Explain. Preview. Verify. Commit.

GUARD

when is change valid?

RULE OBJECT

LINEAGE

remember exactly what changed

Concept profile	DB-0.2
Document type	Game concept + all-ages curriculum + physical-play scaffolding
Audience	Children, families, educators, designers and makers
Prepared for	Tom Klootwijk
Date	28 August 2026

Evidence boundary. This document is an original application design inferred from the supplied UGTS reports. The reports do not claim that *Definition Breaker* is already implemented.

HIGH-LEVEL PRODUCT DECISION

Executive Decision

Make the substrate's most unusual capability both the player's main verb and the learner's main question:

Definitions, operation order, compatibility, guards and transitions become carryable, connectable world objects. The player can propose radical changes, but the world changes only after a deterministic verification and commit.

DECISION

GO for a two-track product: a digital puzzle-adventure plus a physical learning lab.

The digital game proves the systemic fantasy. The physical and print-and-play path lets children encounter the same concepts through stories, matching, routes, gates and replay before they meet formal notation.

GENRE

Systemic puzzle-adventure + construction learning lab

PRIMARY FORMAT

Offline HTML5 + print-and-play + non-electronic toy prototypes

SIGNATURE MECHANIC

Physically rewire typed rule objects and commit verified world patches

CORE EMOTIONAL ARC

Confusion -> understanding -> controlled transgression -> mastery

MVP PROMISE

A 7-room digital slice, one starter kit and four guided lessons

Why it is cool

Instead of finding a key, the player can change what "key," "door," "open," or "before" means - while the game visibly proves whether the edit is legal.

Why it is playable

The player manipulates a bounded subgraph with typed sockets, budgets, preview and readable rejection reasons. It is tactile puzzle solving, not arbitrary coding.

Why UGTS fits

The source line already separates definitions, instances, support, compatibility, guards, commits, patches and lineage, and 3.9 supplies the practical game shell.

Source-supported foundation. Content-addressed definitions and explicit operation order come from 3.6; event-authoritative proposal/commit and replay from Two Hands 3.0; vector gameplay, input, collision, audio and offline export from 3.9.

[S3 pp. 5-13](#)
[S4 pp. 3-13](#)
[S6 pp. 3-15](#)

SOURCE SCREEN AND EVIDENCE DISCIPLINE

Suitable Substrate Versions

The game uses the versions below according to their actual documented strengths. “Core” means required for the first playable; “Optional” means a later learning, editor, metagame or evidence module. Age labels are design entry points, not developmental claims.

Source	Status	Contribution to Definition Breaker	Design boundary
UGTS-GN 1.1	CORE	Canonical support -> compatibility -> guard -> event -> transition -> lineage order; narrow one-bit semantics; precision and memory rules.	Packed flags never become complete state; performance requires equal-error measurement.
UGTS-KC 2.0	CORE	Pattern/field rules, route identity, winding, hybrid automata, event degeneracy and equal-time conflict concepts.	Topology is explicit routing, not physical magic; numerical status stays visible.
KC Two Hands 3.0	CORE	Stable nodes/assets, physical manipulation model, proposal verification, deterministic commit, checkpoints, replay and reason codes.	VR, general physics and live networking remain later work.
UGTS-KC 3.6	CORE	Definitions as first-class data, explicit dependencies, SHA-256 content addresses, operation order, definition queries and typed hinges.	References must resolve; cycles and hidden state are rejected unless a puzzle explicitly models them.
UGTS-KC 3.6.2 SCLP	CORE+	Winding, reflective wrapping, interval guards, bounded grammar, finite seed/index codecs and constraint-release puzzles.	A 64-bit key is an index, not a whole world or infinite detail.
UGTS-KC 3.9	CORE	Vector assets, multi-device input, fixed-step game world, collision, animation, tilemaps, procedural audio, saves and self-contained HTML5.	Native 3D, skeletal animation, multiplayer and general rigid-body physics are not claimed.
3.9.3 IQ Field	OPTIONAL	Rich 2D/3D field primitives, CSG, domain repetition and bounded procedural signals for later rule volumes and boss arenas.	Iterative queries require explicit budgets and status; presentation is never authority.
KC Atlas / 4.0 ledger	OPTIONAL	Evidence-bearing rule revisions, unknown states, proof trails and community-puzzle audit bundles.	Useful for workshop publishing, not required for moment-to-moment play.
3.6.3 SARA	OPTIONAL 13+	Public-versus-secret typing, checksum-as-integrity, explicit authorization and public-only evidence become a digital-citizenship lesson.	No real wallet secrets, key search, signing, balances, transaction features or ownership claims enter the game or toy line.



DEFINITION BREAKER

A SPIRAL CURRICULUM, NOT A SIMPLIFIED DEAD END

How to Use the All-Ages Edition

EXPLORER TRACK

Story, shapes and cause-and-effect

Suggested entry: roughly ages 7-9, families and first-time players.

Read the large plain-language boxes, act out the physical activities and use only the visible rule icons. Technical names are optional.

MATCH -> ORDER -> PREDICT -> EXPLAIN

BUILDER TRACK

Connections, routes and replay

Suggested entry: roughly ages 10-12 or experienced younger learners.

Add typed sockets, dependency chains, winding counters, guard gates, challenge seeds and simple replay records.

CONNECT -> TEST -> REPAIR -> REPLAY

DESIGNER TRACK

Definitions, evidence and authoring

Suggested entry: roughly ages 13+, teens and adults.

Use the formal vocabulary, inspect source boundaries, reason about uncertainty and author bounded puzzle definitions.

MODEL -> VERIFY -> COMMIT -> AUDIT

One book, two directions

Concrete to abstract

Begin with a mat, tile, gate or track. Name what the object does. Then reveal the digital record and finally the formal substrate term.

TOY

support
mat

GAME

room-
only
rule

FORMAL

support
predicate

Abstract back to concrete

Older learners can start with a formal page, then design a physical demonstration that proves they understand it without repeating jargon.

FORMAL

lineage

GAME

replay
trail

TOY

event
cards

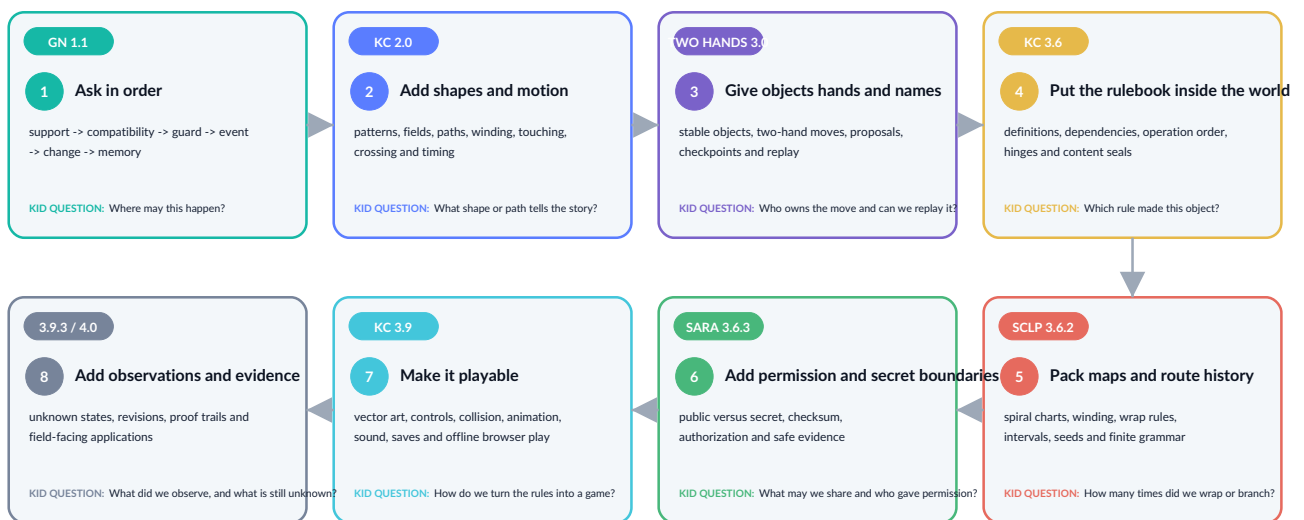
Reading rule. The child-facing pages are an educational interpretation of the source documents. They preserve the documented authority order and boundaries, but stories, age tracks, toy pieces, lesson plans and classroom routines are original scaffolding.

THE HISTORY IS A STAIRCASE, NOT A PILE OF UNRELATED INVENTIONS

How the Substrate Came to Be

HOW THE SUBSTRATE GREW

Each version kept the same safety rule, then added a new layer children can meet through play.



THE CHILD STORY

A careful rule checker learned new skills one layer at a time. First it learned where to look and when a change counts. Then it learned shapes, paths, hands, rulebooks, spiral maps, permission and finally how to become a playable world.

THE SOURCE-BASED READING

The recurring architecture remains support -> compatibility -> guard -> verified event -> transition -> lineage. Later releases add pattern/field calculus, scene and interaction layers, literal definitions, SCLP packing/topology, authorization and a practical vector-game runtime without replacing that authority chain.

Version story synthesized from [S1 pp. 2-5](#) [S2 pp. 3-15](#) [S3 pp. 3-13](#) [S4 pp. 1-14](#) [S5 pp. 5-15](#) [S6 pp. 3-15](#) [S9 pp. 1-12](#).

A CHILD-ACCESSIBLE FOUNDATION FOR THE FULL TECHNICAL VOCABULARY

The Six Ideas Beneath Everything

1

Things have names that survive movement

Child language: The blue bridge is still the blue bridge after we move it.

Formal bridge: Stable node identity, definition identity and position are different.

2

Rules are objects with inputs and outputs

Child language: A rule tile tells what it needs and what it can change.

Formal bridge: Typed definition with domain, codomain, parameters and dependencies.

3

Being nearby is not enough

Child language: Two pieces can share a spot but still belong to different teams or sheets.

Formal bridge: Support and compatibility are separate gates.

4

A change needs a real moment

Child language: The door opens when the marble truly crosses the line, not when it almost touches it.

Formal bridge: Guard classification, tolerance, tangency and hysteresis.

5

Trying is different from changing

Child language: We can preview an idea before the world accepts it.

Formal bridge: Proposal, verification, conflict policy and atomic commit.

6

The world remembers why it changed

Child language: Keep a trail of event cards so another player can replay the solution.

Formal bridge: Lineage, checkpoints, state hashes and replay.

The learning goal is not memorizing six technical words. It is learning to ask six careful questions before changing a system.

TRANSLATE WITHOUT ERASING THE FORMAL DISTINCTION

Vocabulary Bridge I: Things, Rules, Places and Identity

THING / INSTANCE

Story: This exact door in this exact room.

Toy: A door pawn with a unique sticker.

Digital: A scene node or entity with a stable ID.

Do not confuse: The door can change its picture or position without becoming a different door.

RULE / DEFINITION

Story: The recipe that says how doors open.

Toy: A rule card used by one or many doors.

Digital: A content-addressed definition referenced by instances.

Do not confuse: Replacing the shared recipe is different from changing one door.

PLACE / COORDINATE

Story: Where the pawn sits on the table.

Toy: A grid square, ring or route slot.

Digital: Position, phase, sheet and local chart coordinates.

Do not confuse: Coordinates help locate a state; they are not full identity.

HISTORY / LINEAGE

Story: The path the pawn took and the choices that made it.

Toy: A trail of event cards or colored beads.

Digital: Ordered transition history, branch data and replay references.

Do not confuse: A short checksum can help detect change but does not replace the full story.

TYPE / ROLE

Story: A key fits a key socket; a movement rule fits a movement slot.

Toy: Shape-coded tabs plus icon and text.

Digital: Domain, codomain, tags and component role.

Do not confuse: Color alone is never the type.

DEPENDENCY

Story: The bridge rule needs a hinge rule before it can work.

Toy: A cable or arrow from one rule tile to another.

Digital: A resolvable edge in a bounded dependency graph.

Do not confuse: An undeclared loop is not automatically clever; it can be an invalid cycle.

THE SAME WORDS WILL RETURN IN THE GAME, TOYS AND SOURCE DOCUMENTS

Vocabulary Bridge II: Gates, Events and Memory

Formal term	Child-facing object	Question the learner asks	Common misconception to prevent
Support	Mat, hoop, cone or room outline	Where may this rule even try to act?	"Global" is not the default; local relevance comes first.
Compatibility	Matching tabs, team badges or sheet tokens	May these two supported things actually work together?	Same location does not guarantee interaction.
Guard	Gate line, balance scale or threshold dial	Did the promised condition truly happen?	Touching, crossing, staying on the line and noisy chatter are different.
Proposal	Ghost arrangement or "try" tray	What would happen if we used this edit?	A preview is not yet the real world.
Commit	Green stamp or one-way latch	Is the edit valid enough to become authoritative?	A player never changes authority by secretly moving pieces.
Transition	State flip card, route arrow or ownership token	What exactly changes at the event?	A visual effect is not automatically the state change.
Constraint	Shackle, clip or maintained connection	Which motion or relation must continue over time?	Persistent contact should not be an infinite stream of "touch" events.
Lineage / replay	Trail cards, beads and before/after photos	Can another learner reproduce the same result and find the first difference?	A hash detects divergence; it is not the whole state or proof of ownership.

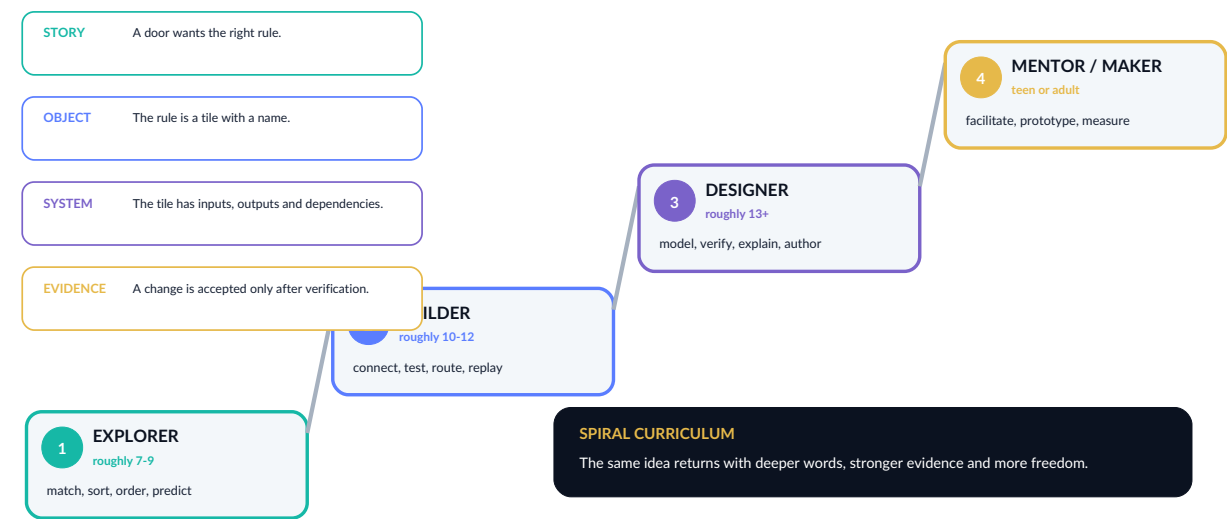
Five-minute family activity. Place two pawns in the same square. Give them different sheet badges. Ask: "Are they close enough?" yes. "Are they compatible?" no. Swap one badge and try again. This single activity introduces support, compatibility and same-place/different-state.

THE SAME CONCEPT RETURNS WITH MORE FREEDOM AND STRONGER EXPLANATIONS

Learning Ladder by Age and Experience

ONE SUBSTRATE, FOUR ENTRY LEVELS

Age bands are suggested entry points, not rigid limits. Learners may move between tracks.



Track	Minimum learner evidence	Optional deeper evidence
Explorer	Can sort rule pieces, predict one outcome and say why a move was accepted or rejected.	Can invent a story that correctly uses “nearby but not compatible.”
Builder	Can repair a short dependency chain, count winding, use a schedule rail and replay a solution.	Can compare two legal orders and explain why the results differ.
Designer	Can specify a bounded rule with inputs, outputs, support, compatibility and guard.	Can document uncertainty, conflict policy, evidence boundary and kill criterion.
Mentor / Maker	Can facilitate without taking over, prototype safely and collect observable learning evidence.	Can translate between physical pieces, digital records and formal source terminology.

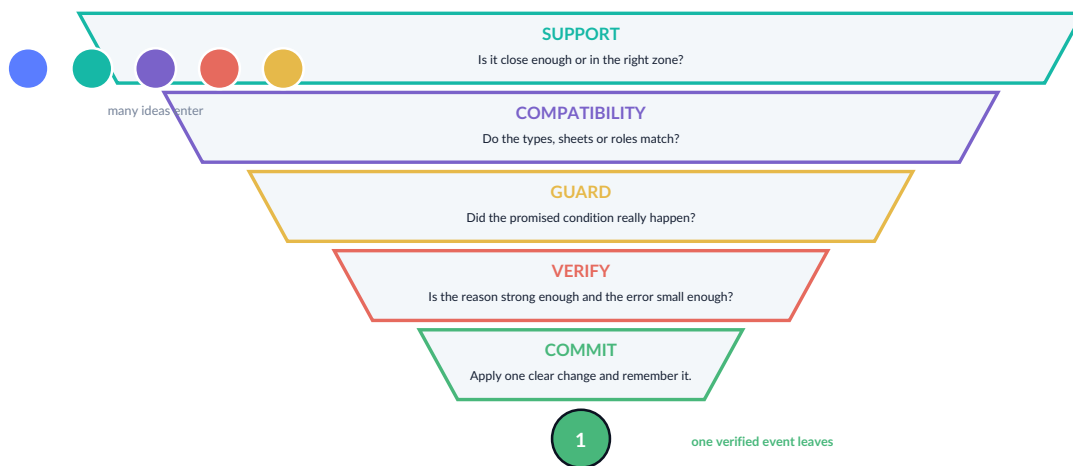
Age boundary. The suggested ages are product-design bands, not assessments of individual development. Literacy, language, sensory access, prior experience and group support matter more than a birthday.

FIRST LESSON: CAREFUL SYSTEMS ASK QUESTIONS IN A STABLE ORDER

UGTS-GN 1.1 for Children: The Great Filter Funnel

THE GREAT FILTER FUNNEL

A change reaches the world only when it passes every question.



Physical game: Pass the Change

1. One learner carries a "change token."
2. A support keeper checks whether the target is on the right mat.
3. A compatibility keeper checks matching role/sheet badges.
4. A guard keeper checks whether the marble crossed the line.
5. A verifier checks the rule card and gives the commit stamp.
6. A recorder adds one lineage card.

FORMAL BRIDGE

The 1.1 addendum makes the canonical query order explicit and insists that one-bit fields have narrow roles. A yes/no token may record support or route selection, but it never contains continuous state, uncertainty, identity and history by itself.

Say this: "The bit answers one question."
Not this: "The bit is the whole world."

Source foundation: [S1 pp. 2-5, 9-15](#).

SHAPES BECOME QUESTIONS, NOT JUST PICTURES

UGTS-KC 2.0 for Children: The Shape Garden

STORY

The garden grows roses, spirals, rolling-circle paths and hidden field hills. Two paths may cross on the page without the travelers meeting, because route history and timing still matter.

Hands-on stations

Pattern stencil table

Trace a rose, spiral, Lissajous or clothoid and compare how the turning changes.

Yarn route crossing

Cross two colored strings. Add time, sheet and route cards before deciding whether an event occurs.

Cross, touch or glide

Move a wheel across a tape line, stop on it, or skim along it. Name the event class before applying a change.

Field hill

Use a height map or magnetic-free peg surface to represent positive, zero and negative field regions.

What the learner is really meeting

Pattern	A versioned curve or shape with parameters and a declared role.
Field	A value everywhere; zero may become a candidate boundary without automatically being exact distance.
Route class	Position plus winding, obstacle history, branch or sheet.
Moving frame	How "forward," "side" and "up" travel along a path.
Hybrid event	Continuous motion plus a discrete guarded change.
Degeneracy	Crossing, touch, tangency, coincidence or simultaneous events need different policies.

Design challenge: Build a route where two pawns occupy the same marked point at different times. Then change only the timing rule so they meet.

Source foundation: [S2 pp. 4-15](#) .

THE PRODUCTION LAYER TURNS FORMAL STATE INTO TOUCHABLE INTERACTION

KC Two Hands 3.0 for Children: Two Hands, One Shared Object

MIDPOINT

Two learners hold a soft beam. The center between their hands tells where the shared object moves.

1

DISTANCE + DIRECTION

Moving hands apart changes scale; turning the line between hands changes alignment.

2

OWNERSHIP + HANDOVER

One learner can join, both can hold, and one can release while object identity stays the same.

3

Physical activity: Cooperative Bridge

1. Place a labeled beam between two supports.
2. Two learners grip marked hand zones.
3. Move, scale and rotate together without changing the beam ID.
4. A third learner records a proposal card.
5. The group checks the target support, compatibility badge and guard zone.
6. Commit or reject, then replay the move from the record.

Important: The physical activity demonstrates interaction structure. It is not a measurement-accurate motion controller or safety-certified load-bearing toy.

Formal concepts behind the play

- Stable scene-node identity versus reusable asset identity.
- Typed left/right hand state and confidence.
- Pinch or grab hysteresis.
- Bimanual translation, separation-based scale, pair-axis alignment and twist.
- Cooperative ownership, join and handover events.
- Proposal verification, deterministic commit, checkpoints, rollback and divergence detection.

Replay detective. One team secretly changes one step in the recorded sequence. The other team replays until the first divergence and names it.

Source foundation: [S3 pp. 5-13, 15-19](#).

THE SUBSTRATE BECOMES SELF-DESCRIBING AND INSPECTABLE

UGTS-KC 3.6 for Children: The Rule Library Inside the World

The door does not just have a rule. The rule has its own name, type, dependencies, place in the schedule and tamper-evident content seal.

Rule-card anatomy

ID	rule:door-open-v3
NEEDS	key-match, door-support, angle-guard
INPUT	door + key + current state
OUTPUT	candidate open patch
PHASE	guard, then transition
SEAL	content fingerprint

Children may start with only the icon and title. Older learners progressively reveal inputs, outputs, dependencies, phases and the canonical content fingerprint.

Three physical lessons

- Dependency train

Connect cards in required order. An unresolved reference leaves an empty carriage.
- Connector hinge

Insert a zero-value connector that changes the graph's structure without changing the numeric total.
- Order rail

Compare MOVE-THEN-ROTATE with ROTATE-THEN-MOVE. Both tiles are legal; their order is not interchangeable.
- Content seal

Change one parameter and compare the new fingerprint. Explain that the seal detects content change, not legal ownership.
- Repair challenge: A lamp depends on a battery, the battery depends on the lamp, and the system rejects the cycle. Add a latch or external start event that breaks the unresolved loop.

Source foundation: [S4 pp. 1-14](#).

COMPACT DOES NOT MEAN MYSTERIOUS, INFINITE OR COMPLETE

SCLP 3.6.2 for Children: Spiral Maps, Wrap Worlds and Seeds

Spiral map

Equal steps on a log-radius chart make multiplicative rings in ordinary space. Children walk from shell to shell and notice that the real distances change.

Boundary: the center still needs a special core region.

Phase + winding

A bead returns to the same clock position every lap, while a separate counter remembers how many laps occurred.

Boundary: repeating phase is not the same as repeating identity.

Reflective wrap

A track seam sends the traveler to another side and flips an orientation token.

Boundary: the route rule is explicit; it is not physical magic.

Guard interval

A value comes with a small “maybe band.” Commit only when the whole band stays safely on one side of the threshold.

Boundary: jitter may not change the event class.

Missing shackle

Remove one declared connector and gain exactly one new legal direction in the toy mechanism.

Boundary: more freedom is not automatically chaos.

Seed + bounded grammar

A short challenge code chooses branches in a finite construction tree. Depth, symbols and stack pieces are visibly budgeted.

Boundary: a 64-bit address is an index, not the whole world.

Capstone activity. Build a three-level branching tree, record one branch bit at each split, place the result on a spiral support mat, cross a reflective seam and add a winding bead. Another team reconstructs the same object from the seed card and rule deck.

Source foundation: [S5 pp. 5-15](#).

A SAFE AUTHORIZATION LESSON WITHOUT IMPORTING WALLET ATTACK MECHANICS

SARA 3.6.3 for Older Learners: Public, Secret and Permission

SAFE STORY

A museum has public exhibit labels, private staff codes and an authorization desk. A label can be checked for typing mistakes, but it does not prove who owns the exhibit or grant permission to open a locked cabinet.

Non-financial learning pieces

- **Public card:** safe information designed to be shown.
- **Secret envelope:** a placeholder that is never opened, copied or entered into the game.
- **Checksum strip:** catches some accidental changes but does not authenticate a person.
- **Permission badge:** a requested activity proceeds only when scope is allowed.
- **Public-only certificate:** records what was checked without leaking secrets.

Source-grounded lesson

Public and secret are different types	Do not collapse an address, key, seed, authorization and ownership into one object.
Integrity is not identity	A valid checksum means the public string is structurally valid, not that a person owns it.
Permission is a guard	Allowed, self-owned or test-network scope must be explicit before an audit proceeds.
Secrets do not enter lineage	Logs and certificates stay public-only.

Non-negotiable classroom boundary. Use fictional museum, diary or message examples. Do not collect real wallet words, passwords, private keys, recovery phrases, account secrets or financial identifiers. The game exposes no search, signing, network, balance or transaction path.

Source foundation: [S9 pp. 1-12](#) . Suggested only for older learners and supervised digital-citizenship contexts.

THE RUNTIME IS THE PLAYGROUND, NOT THE AUTHORITY SHORTCUT

KC Elizabeth 3.9 for Children: Turning Rules into a Playable World

ART**Vector world**

Paths, fills, strokes and gradients stay sharp and inspectable.

INPUT**Many ways to play**

Keyboard, gamepad, pointer and touch map to named actions.

TIME**Fixed steps**

The world advances in an explicit order so replay and testing are meaningful.

WORLD**Collision and gameplay**

Players, bodies, sensors, hazards, collectibles, bounds and camera create the room.

FEEDBACK**Animation and sound**

Motion, particles and procedural audio explain acceptance, rejection and success.

SAVE**Remember and share**

Snapshots, hashes, project records and offline HTML5 turn lessons into playable artifacts.

Child-facing digital modes

- **Guided mode:** one idea and a small visible rule set.
- **Build mode:** snap rule objects into authored sockets.
- **Trace mode:** show the ordered questions and rejection reason.
- **Replay mode:** compare two attempts and find the first divergence.
- **Workshop mode:** older learners author bounded puzzles without arbitrary scripts.

ONE PROJECT, TWO TARGETS

The same canonical project can feed headless simulation and a self-contained browser build. That makes it possible to test a child's solution record, then present it through accessible vector graphics, touch controls and procedural feedback without changing the underlying rule data.

Source foundation: [S6 pp. 3-15](#).

CHILDREN LEARN TO DISTINGUISH OBSERVATION FROM CONCLUSION

Evidence Lab: Observe, Mark Unknown, Revise

A careful learner may say “I do not know yet.” Unknown is a valid state, not a failure to be hidden.

Evidence cards

OBSERVATION	“The red route piece moved two squares.”
INTERPRETATION	“The bridge rule may have changed.”
CONFIDENCE	high / medium / low / unknown
REVISION	new evidence updates a previous claim without erasing its history

Optional later-profile bridge: **S8**.

Workshop use

1. Photograph or diagram the starting arrangement without personal data.
2. Record what was directly observed.
3. Separate the proposed explanation from the observation.
4. Attach a confidence or unknown marker.
5. Commit only a supported revision.
6. Keep the older record so another learner can audit the change.

Route evidence game. One team blocks a route. Another team receives incomplete observations and must decide whether the route is reachable, unreachable or still unknown. A safe answer may be “insufficient evidence.”

Boundary. Toy and classroom evidence is not a structural-safety, medical, emergency-navigation or legal verdict. The lesson is epistemic discipline: what was observed, what was inferred and what remains unknown.

A SHARED LANGUAGE FOR CHILDREN, EDUCATORS, GAME DESIGNERS AND ENGINEERS

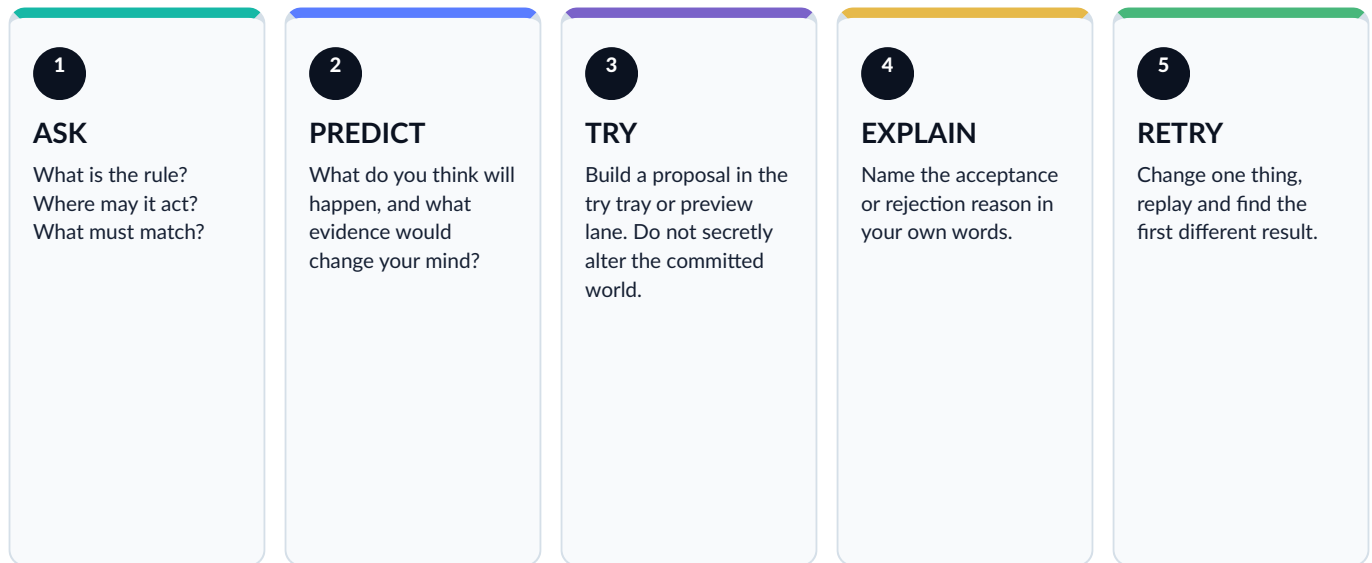
Physical-to-Digital Concept Crosswalk

Core concept	Physical learning object	Definition Breaker mechanic	Formal substrate representation
Stable identity	Uniquely labeled pawn or beam	The object survives movement, reskinning and handover	node_id / entity_id distinct from asset and position
Definition	Reusable rule tile	One rule can control many instances	content-addressed definition record
Support	Mat, hoop, cone or room outline	Rule acts only in admitted zones	support predicate / bounds
Compatibility	Matching tab, badge or sheet token	Supported objects couple only when tags match	typed compatibility predicate
Guard	Gate line, dial or balance	Change triggers at a verified condition	relation, interval, accepted status and margin
Order	Schedule rail	Players reorder legal operators	phase-ordered pipeline
Topology	Sheet overlay, portal track, winding bead	Route, orientation and sheet change explicitly	gluing map + state patch + lineage
Constraint	Clip, shackle or hinge	Maintain or release a relation	constraint mode / row release / freedom gain
Proposal and commit	Try tray + green stamp	Ghost preview, verify, atomic world change	EventProposal -> RuntimeEvent
Lineage and replay	Event cards, beads and before/after state	Solutions are reproducible and divergence is visible	pre/post hashes, checkpoints and ordered events
Seed and grammar	Challenge code + branch cards	Reconstruct bounded room variants	seed, finite grammar and compact index
Authorization	Permission badge + sealed envelope	Older-learning safety gate	allowed scope, secret/public partition, reason codes

Design rule. Every toy or child-facing metaphor must preserve at least one formal distinction and explicitly name what it does not represent. Metaphor is a bridge, not a license to collapse terms.

THE ADULT PROTECTS THE QUESTION, NOT THE ANSWER

Facilitator Guide: Ask, Test, Explain, Retry



Useful facilitator prompts

- “What stayed the same even though the object moved?”
- “Are these pieces nearby, compatible, both, or neither?”
- “Did the marble cross, touch, stay, or chatter?”
- “Which rule must happen first?”
- “What would make this proposal safer or clearer?”
- “Can another team replay your result?”
- “What do we know, what do we think, and what is unknown?”

Group roles



Do not over-help

Reveal one missing question rather than the solution. For example: “Have you checked compatibility?” is better than “Swap the purple token.” A rejection reason is part of the learning material.

Accessibility. Read cards aloud, allow pointing instead of speaking, provide tactile shapes, never rely on color alone, permit extra planning time and offer cooperative rather than speed-based scoring.

LEARNING IS VISIBLE IN THE TRAIL, NOT ONLY THE ANSWER

Learning Evidence, Portfolios and Badges

Portfolio packet for one challenge

1. **Starting world:** photo, drawing or piece list.
2. **Prediction:** spoken, drawn or written.
3. **Proposal:** the attempted rule arrangement.
4. **Verifier result:** accepted or one exact rejection reason.
5. **Revision:** what changed in the second attempt.
6. **Committed world:** final arrangement and lineage trail.
7. **Replay check:** another learner reproduces or finds divergence.
8. **Reflection:** one formal term the learner can now explain with an example.

Privacy. Portfolios should use first names or classroom aliases only. Do not include dates of birth, government identifiers, passwords, financial details, wallet words or unrelated personal information.

Evidence-based badge ladder

SORTER

Correctly groups pieces by role or type.

ROUTE READER

Explains same-place/different-path or winding.

RULE REPAIRER

Fixes a dependency, socket or order problem.

GUARD KEEPER

Distinguishes crossing, touch, uncertainty and chatter.

PREVIEW PILOT

Uses a failed proposal as information.

REPLAY DETECTIVE

Reproduces a solution and finds first divergence.

SYSTEM DESIGNER

Authors a bounded challenge with evidence boundary and kill criterion.

Assessment principle. Reward explanations, revisions and reproducibility - not only speed or a correct final arrangement.

WORLD, ROLE AND PROMISE

High Concept and Player Fantasy

The premise

The player is a **Breaker**, a maintenance operative inside the Definition Stack: a city whose architecture is compiled from visible rules. After an event called **The Unresolved Cycle**, definitions detach from their instances. Doors inherit gravity, corridors forget which sheet they occupy, alarms and bridges commit contradictory patches, and some rooms depend on themselves.

The Breaker carries a Rule Lens and a portable Commit Bench. They can pull authorized rule objects from sockets, route typed cables, reorder phases, insert hinges, change topology ports and remove constraints. But raw manipulation is only a proposal. The world accepts a change after the substrate proves the edit is well-typed, supported, compatible, numerically safe, invariant-preserving and conflict-resolved.

“You are allowed to break the rule only after you understand exactly what the rule is.”

Player fantasy

- See the hidden logic that makes a world behave.
- Touch and rearrange that logic as tangible machinery.
- Use rejection, replay and lineage as investigative tools.
- Turn an impossible room into a valid new world - without cheating the authority model.

UNIQUE SELLING PROPOSITION

The laws of the level are inventory items.

Definitions are not abstract menu entries. They have mass, sockets, phase labels, dependencies, seals and history. Carrying a rule from one machine to another can be as meaningful as moving a crate.

Not a coding game

Players do not write arbitrary text. They manipulate a constrained, visual and tactile rule graph with authored choices, typed connectors, budgets, previews and undo.

Not pure puzzle-menu play

The changed definitions immediately alter traversal, hazards, NPC behavior, topology, timing and object ownership in the physical room.

Not consequence-free reality editing

Every accepted change leaves a lineage record. Later rooms may inspect what the player changed, in which order, and under which policy.

Game-design inference. The narrative, Breaker role and city are original design choices. Their mechanical credibility comes from the substrate's separation of definitions, instances, verified events, patches and lineage. [S4 pp. 3-6](#) [S3 pp. 6, 11-13](#)

THE GAME IN ONE CYCLE

Player Verbs and Core Loop



Inspect

Reveal stable IDs, type signatures, dependencies, operation phase, compatibility tags, guards, invariants and lineage.

Manipulate

Carry, socket, cable, rotate, reorder, branch, gate, hinge, wrap, lock, release or quarantine a bounded rule object.

Reason

Scrub a deterministic preview. Compare current and proposed state. Read rejection reasons as clues.

Exploit

Commit the valid patch, traverse the altered room, evade hazards, recover definitions and preserve the needed lineage.

Moment-to-moment rhythm

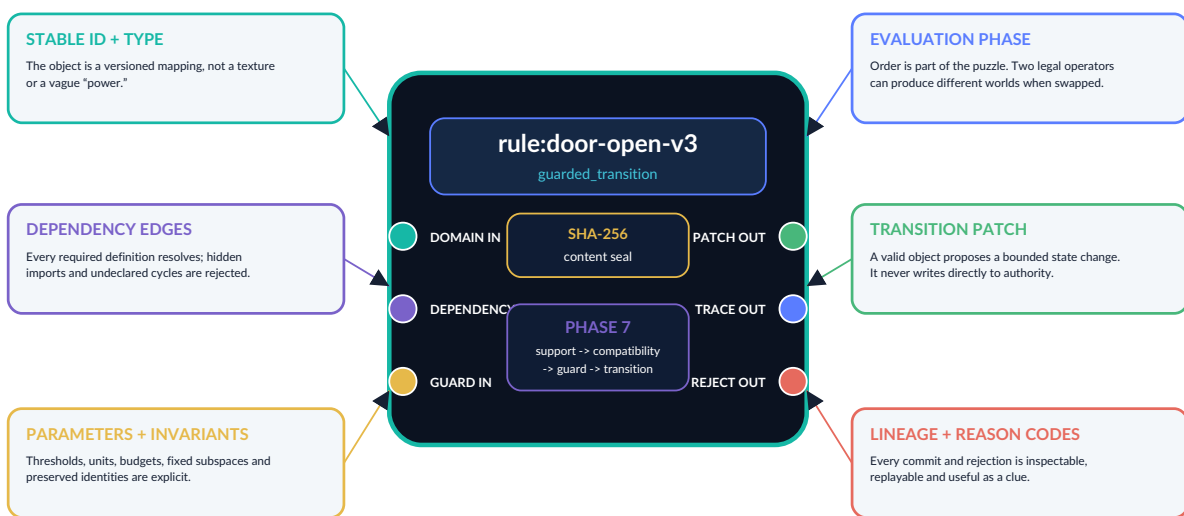
1. Enter a room and discover the physical symptom.
2. Use the Rule Lens to expose the relevant subgraph.
3. Find or earn access to a small set of rule objects.
4. Build a candidate edit on the Commit Bench.
5. Preview consequences and inspect trace/rejection.
6. Commit, then execute the traversal or timing challenge.

What produces mastery

- Recognizing type and dependency patterns.
- Predicting non-commuting operation order.
- Distinguishing coordinates from identity.
- Understanding crossing, touch, tangency and persistent constraints.
- Planning around deterministic conflict resolution and replay.

DATA MODEL TRANSLATED INTO GAME LANGUAGE

A Rule Object Is a Physical Definition



Non-negotiable design rule. The physical prop is a presentation of a canonical definition record. Picking it up does not change its identity; editing its parameters creates a versioned definition or a patch. A visual appearance, cable or icon never silently becomes the authoritative semantics. [S4 pp. 3-7](#) [S3 pp. 5-8](#)

WHAT THE PLAYER CAN CARRY

Rule Object Taxonomy

Literal Tile

Number, vector, tag, threshold, duration, phase, sheet or policy value. Small, stackable and easy to duplicate when allowed.

Typical puzzle: choose the right threshold without shrinking the event margin below safety.

Operator Core

A typed mapping with domain, codomain, parameters and capabilities: transform, compare, combine, project, integrate, route.

Typical puzzle: substitute an operator whose output actually matches the next socket.

Hinge

Continuous pivot, discrete parity selector or zero-magnitude connector with nonzero structural role.

Typical puzzle: insert a connector that changes graph order without changing numeric value.

Support Volume

Local admission region such as sphere, cone, shell, tile region or finite field support.

Typical puzzle: prevent a global rule from affecting unrelated objects.

Compatibility Seal

Sheet, phase, mode, orientation, ownership and application tags that determine whether supported states may couple.

Typical puzzle: make two co-located systems genuinely compatible.

Guard Blade

Boundary, threshold or relation with accepted statuses, confidence floor, error interval and hysteresis.

Typical puzzle: distinguish a true crossing from a tangency or chatter.

Transition Key

An atomic state, route, topology or dynamic reset applied only after verification.

Typical puzzle: change the requested patch without breaking stable node identity.

Topology Port

Explicit gluing, wrap, orientation transport, sheet update and winding lineage.

Typical puzzle: select a half-turn bundle or a genuinely reflective quotient.

Schedule Rail

Stores the actual evaluation/composition order. Blocks on the rail do not commute unless their definitions say so.

Typical puzzle: reorder translate, rotate, gate and commit.

Constraint Shackle

A maintained relation over an interval; removing one declared row may add a legal degree of freedom.

Typical puzzle: release exactly one motion without turning contact into infinite events.

Lineage Seal

Pre/post state hashes, committed event, source, branch and history reference.

Typical puzzle: preserve a required route history while changing position.

Quarantine Cage

Contains unresolved references, invalid hashes, forbidden cycles, overflowed grammar or incompatible definitions.

Typical puzzle: repair the graph rather than bypass validation.

Presentation inference. Names such as “Guard Blade” and “Lineage Seal” are game-language skins over source-supported record classes; the underlying semantics stay typed and inspectable.

MAKE FORMAL STATE READABLE AND TACTILE

The Commit Bench and Rule Lens

Rule Lens: understand the current world

RULE LENS / TARGET: bridge_17

Instance	node:bridge_17
Definition	geom:hinged_bridge-v2
Phase	4 transform
Support	room_03 only
Conflict	translate precedes rotate

parse -> resolve -> construct ->
transform -> support ->
compatibility -> guard ->
transition -> lineage

The lens has three layers: **simple** (icons and plain language), **trace** (ordered operations and reasons) and **definition** (full typed record for expert players and workshop authors).

Physical input

- **Mouse/touch:** drag blocks, cable sockets, scrub preview, long-press to inspect.
- **Gamepad:** radial selection, snap-to-socket, shoulder-button phase movement.
- **Two-hand future:** stretch the bench, align ports and twist a rule cluster using the 3.0 bimanual transform.

Commit Bench: edit a candidate world

CANDIDATE PATCH / NOT AUTHORITATIVE

3 CONSTRUCT	bridge
4 TRANSFORM	rotate 90 -> translate +2
5 SUPPORT	room_03
6 COMPAT	player + bridge
7 GUARD	angle >= 90

PREVIEW HASH: 3bd1...

VERIFYCOMMIT

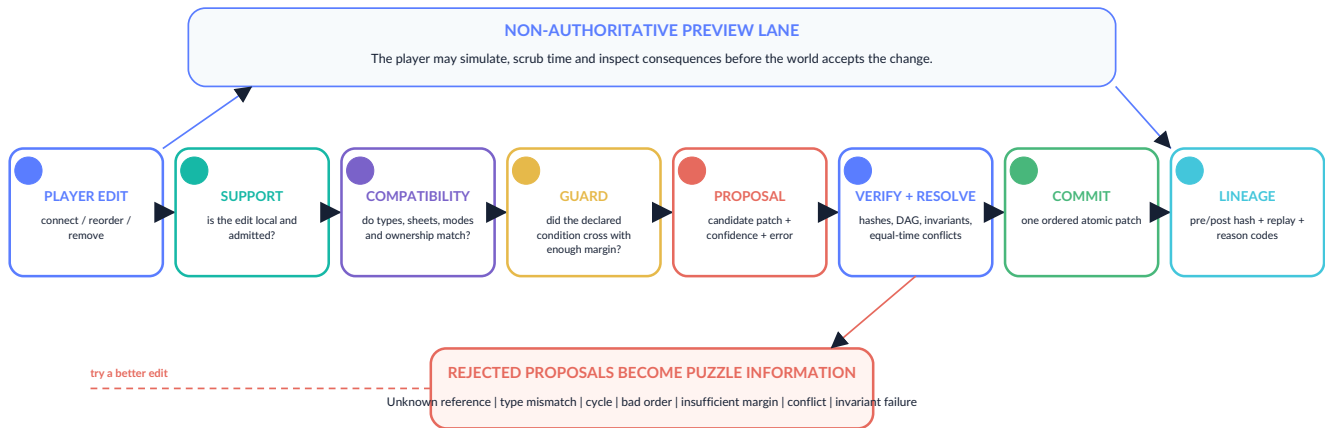
Usability protections

- Only sockets relevant to the current puzzle are exposed by default.
- Every block has shape, icon and text coding; color is never the only cue.
- Invalid cables refuse to snap and explain the domain/codomain mismatch.
- Preview is visually ghosted; the committed world is never ambiguous.
- Undo is checkpoint/replay, not a fragile reverse mutation.
- A hint can reveal one missing gate or one conflicting dependency without solving the entire graph.

Interaction boundary. The 3.9 action map supports keyboard, gamepad, pointer and touch. The richer two-hand manipulation is a later adapter target; the first release should not depend on native OpenXR. [S6 p. 6](#) [S3 pp. 11-12, 17](#)

THE GAME'S GOVERNING LAW

Every Edit Is a Proposal, Not a Cheat



Verification checklist

1. Definition IDs resolve and content hashes verify.
2. Dependency graph is acyclic under the selected cycle policy.
3. Socket domains/codomains and units match.
4. Edit lies inside an admitted support zone.
5. Sheet, phase, mode, orientation and ownership are compatible.
6. Guard status is accepted; numerical error is within margin.
7. Declared invariants and stable identities survive.
8. Same-time patch conflicts are ordered, merged or rejected.

Why this is fun rather than bureaucracy

The verifier is the puzzle's "physics." A rejection is concrete information: the player learns that a route is on the wrong sheet, a dependency is cyclic, a crossing is only a touch, or two valid edits conflict at the same tick.

Later levels deliberately require using the verifier as an instrument: construct a failed proposal to reveal a hidden phase, then repair it.

The pipeline preserves the recurring UGTS authority chain and the 3.0 proposal/commit split.

S1 pp. 2-4, 28

S2 pp. 11-12, 15

S3 pp. 5, 13

FAILURE BECOMES LEGIBLE

Preview, Rejection and Deterministic Explanation

Current world

The last committed state. Solid colors, collision and gameplay authority are active. Its canonical hash anchors saves, replay and puzzle goals.

CURRENT / 8ef1...

Candidate preview

A sandboxed simulation of the proposed graph. Ghosted geometry, predicted routes, contact traces and potential conflicts are visible, but nothing mutates authority.

PREVIEW / 3bd1...

Committed world

The verified patch is applied atomically; lineage and a replay record are appended. The new world receives a fresh state hash.

COMMITTED / a24c...

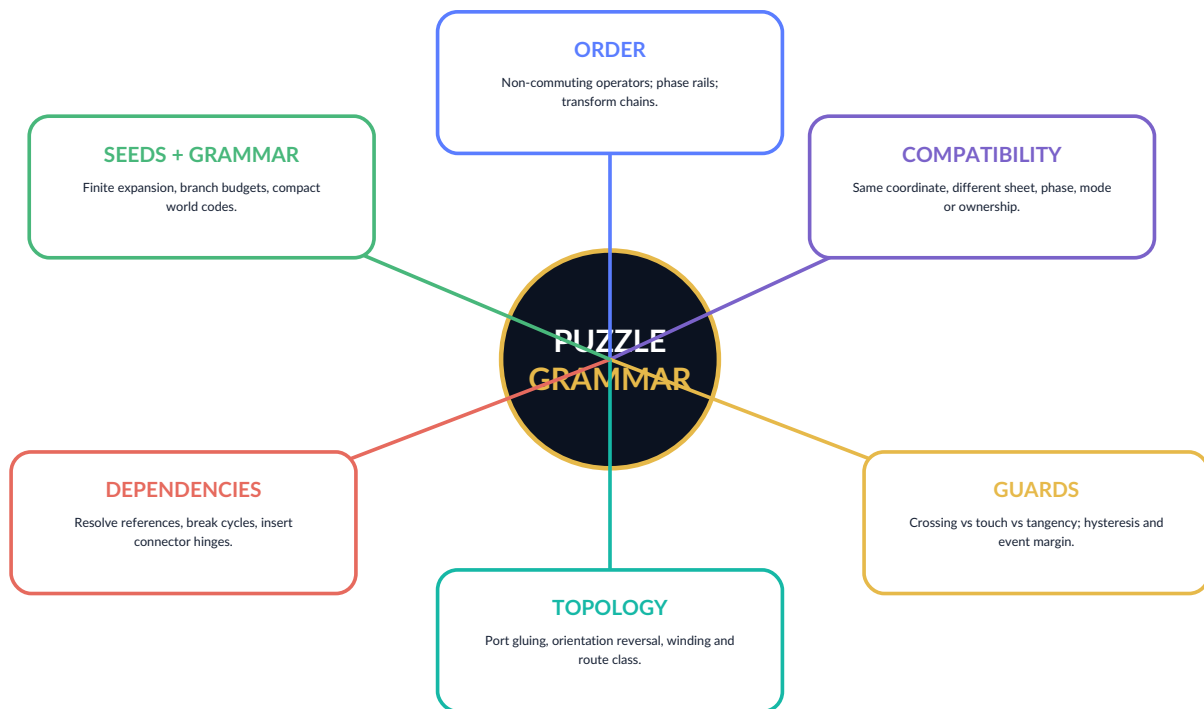
Rejection is a designed feedback channel

Reason code	Player-facing message	Puzzle meaning
unknown_reference	"This rule points to something that does not exist here."	Find the missing definition or replace the dependency.
type_mismatch	"A route is not a velocity."	Insert a projector/adaptor or use the correct socket.
dependency_cycle	"The door waits for the power that waits for the door."	Break the cycle with a latch, initial literal or connector hinge.
incompatible_state	"These objects overlap, but they do not occupy the same state sector."	Resolve sheet, phase, mode, orientation or ownership.
guard_ambiguous	"The relation interval touches both sides of the boundary."	Increase margin, improve the measurement or choose a safer guard.
patch_conflict	"Two valid rules request different values at the same event time."	Set priority, prove commutativity or prevent one proposal.
invariant_failure	"The edit would erase a stable identity or break topology."	Change the patch, not the identity; preserve required incidence/gluing.
budget_overflow	"This grammar would grow beyond the room's declared limit."	Reduce depth/branching or reuse a shared definition.

Determinism boundary. The design requires stable ordering and hash-checked replay in the selected runtime profile. It does not assume universal bit-identical floating-point behavior across every future CPU/GPU. [S3 p. 13](#)

REUSABLE FAMILIES FOR A FULL CAMPAIGN

The Six-Family Puzzle Grammar



Authored freedom, not arbitrary freedom

Each room exposes a deliberately small editable subgraph: usually 3-8 movable rule objects, 2-5 fixed definitions and one explicit goal. The space is rich enough for multiple valid solutions but bounded enough to read and test.

Solutions are proofs

A solution is not merely reaching the exit. The final state must satisfy the room goal, preserve required invariants and reproduce under deterministic replay from the room checkpoint.

THE FIRST HALF OF THE GAME'S LANGUAGE

Puzzle Families I-II: Order and Compatibility

I. ORDER / NON-COMMUTATIVITY

Two individually valid operators can produce different geometry or state when composed in a different order. The player physically moves blocks along a phase rail or rotates a transform chain.

Core patterns

- Rotate then translate versus translate then rotate.
- Apply a field morph before or after support clipping.
- Evaluate a hinge before a topology wrap so chirality is transported correctly.
- Resolve a guard before a dynamic reset rather than after it.
- Place a connector hinge that changes graph order while carrying zero numeric magnitude.

Readable cues

Ghost paths, phase-colored trails, before/after pivots, and a “first divergence” marker show where two traces become different.

Mastery challenge: construct two different valid orders that reach the same room goal but leave different lineage, then choose the one required by a later audit door.

II. SUPPORT / COMPATIBILITY

Spatial overlap is only the first question. Objects may still be separated by sheet, phase, route class, orientation, mode, policy or ownership.

Core patterns

- Two corridors cross in projection but occupy different sheets.
- A sensor sees the crate, but its support cone excludes the correct face.
- A hazard and player share a tile but are incompatible during a phase window.
- A rule is valid only inside a declared room support, preventing global side effects.
- A two-hand object remains owned after handover even though its position is unchanged.

Readable cues

The Rule Lens separates state sectors into layers, shows support volumes as translucent geometry, and marks incompatible overlaps with a dashed “no coupling” seam.

Mastery challenge: preserve two distinct states at one coordinate until a short compatibility window allows a controlled exchange.

Order as literal data and non-commuting hinges: [S4 pp. 5-6, 9, 13](#). Coordinate equality versus identity/compatibility: [S1 pp. 4-5, 18-19](#) [S2 pp. 4-5, 8](#).

TIME AND ROUTE BECOME MANIPULABLE

Puzzle Families III-IV: Guards and Topology

III. GUARD / EVENT TIMING

A value near zero is not automatically a crossing. The player must reason about direction, derivative, interval, hysteresis, priority and whether a relation should become a maintained constraint.

Core patterns

- Crossing, touch, tangency and coincidence produce different outcomes.
- A fast object needs a swept guard rather than a sampled overlap.
- Enter and exit thresholds prevent a switch from chattering.
- A persistent grab/contact must become a constraint mode, not infinite events.
- Two event-time intervals overlap and require priority, merge or conflict.

Signature room: "The Door That Opens Too Late." Move the predictive guard ahead of collision or add a certified swept relation so the bridge commits before the runner arrives.

IV. TOPOLOGY / ROUTE IDENTITY

Portals and wraps are explicit coordinate, orientation, sheet and lineage maps. The player manipulates route classes rather than relying on impossible visual tricks.

Core patterns

- Möbius-style wrap flips orientation after one traversal.
- Reflective Klein gluing is distinct from a simple half-turn bundle.
- Winding count distinguishes repeated phase coordinates.
- A self-crossing path may represent separate branches or route classes.
- Cell-complex edits must preserve incidence invariants.

Signature room: "Klein Freight." Route a marked crate through a reflective wrap, compensate its chirality, and deliver the correct orientation without merging two incompatible sheets.

Event degeneracy and persistent constraints: [S2 pp. 11-12](#) [S3 p. 11](#) . Explicit wrapping, winding and chirality: [S5 pp. 5-10](#) .

THE ADVANCED PUZZLE LANGUAGE

Puzzle Families V-VI: Dependencies and Seeded Grammar

V. DEFINITION GRAPH / REFERENTIAL CLOSURE

Definitions may depend on other definitions, but every reference must resolve, hashes must verify and the graph must obey its cycle policy. The player repairs the actual logic graph behind broken machines.

Core patterns

- Missing references create orphaned machinery.
- Duplicate IDs produce ambiguous sockets.
- A definition hash fails after an illicit parameter edit.
- A door-power-door dependency creates an unresolved cycle.
- A connector or latch breaks the cycle while preserving value and order.
- Replacing a definition changes every instance that references it; replacing one instance does not.

Signature room: "Gravity Is a Dependency." Detach a local room's gravity definition without changing the global definition used by the rest of the station.

VI. SEED / BOUNDED GRAMMAR / COMPRESSION

Rooms and challenge variants can be reconstructed from a seed, finite grammar and compact index, while external novelty and committed edits remain logged.

Core patterns

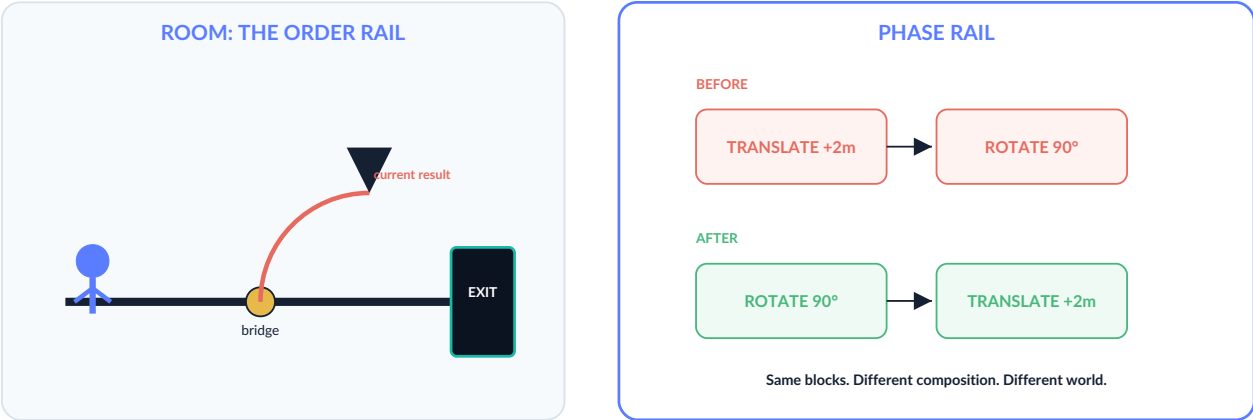
- Branch bits select one of two declared productions.
- Grammar depth, symbols and stack are budgeted.
- A 64-bit address chooses a quantized local cell/phase, not full state.
- Prefix refinement narrows a search region but does not directly halve physical coordinates.
- Seed + definitions reconstruct the baseline; player edits remain event history.

Signature room: "Worldseed 64." A map key locates the missing rule cache, but the player must recover the separate definition and lineage records to make the room whole.

Compression boundary. The game may exploit shared definitions, vector assets, bounded procedural generation and event-log reconstruction. It must never present nominal bit-width ratios as equal-information compression without a reconstruction and error contract. [S5 pp. 7-12](#) [S1 pp. 9-13](#)

DETAILED PLAYABLE SCENARIO 1

Example Room: The Order Rail



Room symptom

A hinged bridge rotates around the wrong point and misses the exit ledge. The same translation and rotation blocks are already present; the machine is “valid” but ordered incorrectly.

Intended discovery

Composition is non-commutative. The Rule Lens marks the first divergence between the current and candidate transform traces.

Solution

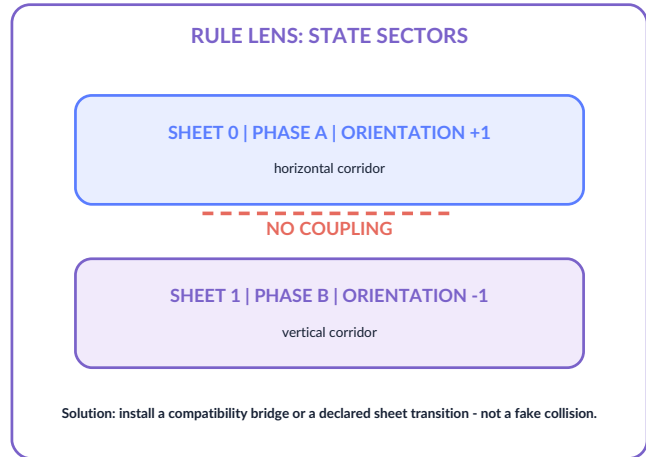
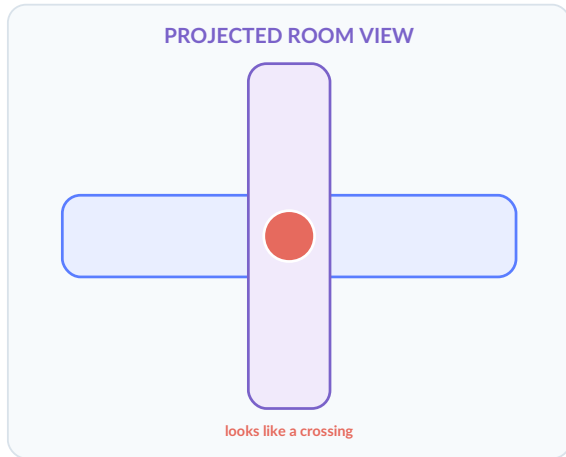
Move **ROTATE 90** before **TRANSLATE +2m** on the phase rail. Preview the pivot path, verify, then commit the transform-order patch.

Available objects	Hidden constraint	Failure feedback	Optional expert solution
Rotate, translate, pivot literal, room support, angle guard.	Bridge node identity and hinge pivot must remain stable.	“Invariant failure: pivot moved with the asset.”	Keep the original order but insert a compensating frame transform; produces a different lineage accepted by the goal.

Teaching purpose. This room turns 3.6's explicit non-commuting composition and 3.0's stable node identity into a readable first-act puzzle. [S4 p. 6](#) [S3 p. 6](#)

DETAILED PLAYABLE SCENARIO 2

Example Room: Same Place, Different Sheet



Room symptom

The player sees a corridor cross and expects to turn upward, but the avatar passes underneath. A power pulse appears to touch a receiver yet never couples.

Available edits

- Compatibility bridge: temporarily match sheet and phase.
- Declared sheet transition: move one route to the other sheet.
- Projection collision: tempting but invalid; it treats appearance as authority.

Intended solution

Install a compatibility window gated by the pulse phase, allowing energy to transfer for exactly one tick while the player routes through a separate portal. The world remains layered; the solution does not collapse two identities into one.

Failure feedback

incompatible_state: "Supported overlap found.
Coupling rejected: sheet 0 / phase A versus sheet 1 / phase B."

Conceptual boundary. The game may visualize sheets as layered corridors, but it should not imply that topology creates physical energy or that projected self-intersection is automatically contact. [S1 pp. 4, 18-19](#) [S2 p. 8](#)

DETAILED PLAYABLE SCENARIOS 3-4

Example Rooms: Missing Shackle and Klein Freight

THE MISSING SHACKLE

A suspended counterweight is constrained in two velocity directions. The exit mechanism needs one new legal degree of freedom, but simply deleting all constraints drops the mass and destroys the timing sequence.

Available objects

- Two constraint rows, one damped hinge, tangent projector, safety guard and reset.
- A “release one row” tool that exposes the nullity change.

Solution structure

1. Remove exactly the lateral constraint row.
2. Keep the radial row and tangent projector.
3. Attach the damped hinge so the arm swings rather than accelerates without bound.
4. Commit only when the safety guard certifies the new path.

Core lesson: constraint release increases legal motion; it does not automatically supply force, chaos or a trajectory.

KLEIN FREIGHT

A cargo crate must traverse a radial wrap. The displayed path looks like a half-turn, but the destination scanner requires a genuinely orientation-reversed state and a preserved winding label.

Available objects

- Source half-turn bundle, reflective Klein wrap, chirality automorphism, orientation seal and winding counter.

Solution structure

1. Select the reflective wrap rather than the orientation-preserving base rotation.
2. Let the wrap flip local orientation and hoop state.
3. Apply the chirality map to the crate's control convention.
4. Preserve the winding in lineage so the scanner distinguishes first and third arrival.

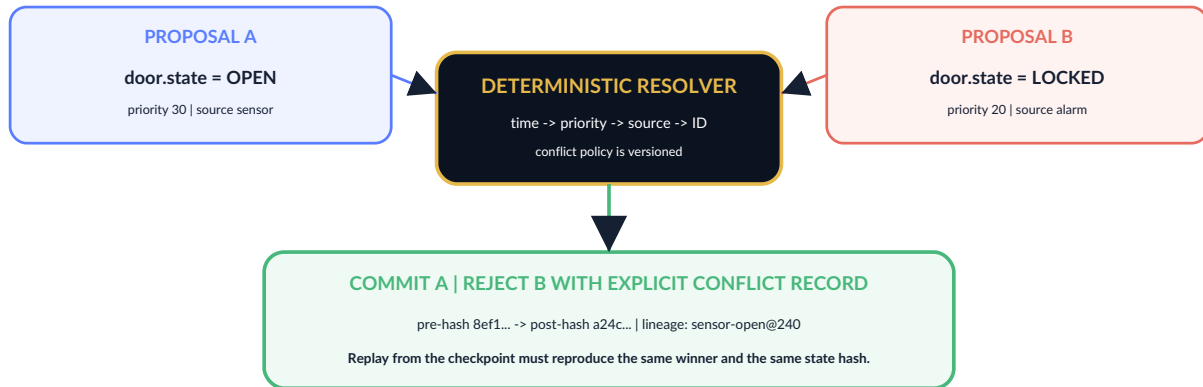
Core lesson: visually similar wraps can have different derivative/orientation maps and therefore different gameplay semantics.

Constraint row release and tangent projection: [S5 pp. 6-7](#). Half-turn versus reflective Klein profile and chirality: [S5 pp. 5-10](#).

DETAILED PLAYABLE SCENARIO 5

Example Room: Conflict at Tick 240

TICK 240: TWO VALID PROPOSALS, ONE CONFLICTING FIELD



Room setup

A sensor proposes **OPEN** while an alarm proposes **LOCKED** at the same fixed tick. Both proposals independently satisfy support, compatibility and guard checks. The door cannot accept both values.

Player task

Trace the previous loop, discover why the alarm fires, then either prevent one proposal, change the declared priority, or replace the patches with a commutative state representation such as an explicit *open-but-alarmed* mode.

Replay reveal

The game rewinds to the checkpoint and replays input and events. A ghost trace shows the same conflict and the same winner. The player can scrub directly to the first hash divergence after an edit.

Expert variant

Two players or two echoes each control one source. They must coordinate timing so the proposals no longer overlap, avoiding a priority hack.

Design inference. Turning deterministic conflict resolution into a time-loop puzzle is an application concept. The underlying ordered proposal, rejection, pre/post hash and replay mechanisms are source-supported. [S2 pp. 11-12, 21](#) [S3 p. 13](#)

[S6 pp. 6, 9, 12](#)

TEACH ONE DEEP DISTINCTION AT A TIME

Campaign Structure and Progression



The campaign increases expressive power, but never removes typing, budgets, preview or deterministic commit.

Chapter	Rule vocabulary	Mechanical escalation	Boss / set-piece
I. The First Hash	Stable ID, literal, type, instance versus definition.	Match sockets; repair one missing reference; learn preview/commit.	The Orphan - a machine with no definition.
II. The Phase Rail	Operation order, hinges, transforms, invariants.	Reorder blocks; preserve pivots; exploit connector hinges.	The Commutator - changes order every loop.
III. False Twins	Support, compatibility, sheet, phase, ownership.	Manage co-location without false coupling; timed compatibility windows.	The False Twin - two identical bodies with different lineage.
IV. The Event Horizon	Guard margin, crossing, touch, tangency, hysteresis, constraint.	Predict events; avoid chatter; convert persistent contact to a maintained mode.	The Chatter Engine - floods the room with invalid zero events.
V. Winding City	Port gluing, orientation, sheet change, winding and route class.	Route cargo, player and signals through different quotient rules.	The One-Sided Station - orientation is the actual health bar.
VI. Replay Court	Proposals, priorities, conflict, checkpoints, hashes, lineage.	Use past loops as proof; construct reproducible multi-event solutions.	The Auditor - accepts only solutions whose replay and lineage match.
VII. Definition Breaker	Dependencies, bounded grammar, seed/index, full authority chain.	Repair the Unresolved Cycle and decide which definitions the city should keep.	The Optimizer - tries to collapse identity, uncertainty and history into one bit.

TURN FORMAL DISTINCTIONS INTO MEMORABLE DRAMA

Narrative, Characters and Boss Logic

The thematic conflict

Definition Breaker is about the difference between freedom and unverified power. The city failed because its administrator, **The Optimizer**, attempted to compress every distinction - identity, state, uncertainty, route history and meaning - into minimal flags. The resulting "efficient" world is unable to explain itself and cannot safely change.

The Breaker restores not a single correct order, but an inspectable order: typed definitions, explicit dependencies, bounded edits and preserved lineage. The final choice can offer multiple valid city constitutions, each replayable and auditable.

Key characters

- **Mira / The Instance:** a recurring NPC whose body and asset change, but whose stable node identity and lineage persist. Teaches identity separation.
- **Latch:** a small connector-hinge companion that has "zero magnitude, nonzero structural role." It becomes a literal puzzle tool and comic voice.
- **The Auditor:** antagonist-turned-ally who refuses undocumented miracles; teaches proof, replay and reason codes.
- **The Orphans:** machines and citizens whose definitions are missing. Restoring them raises ethical questions about version and provenance.

THE COMMUTATOR

Reorders transform and event blocks. Damage is dealt by pinning a deterministic phase order long enough to make its own attacks miss.

THE FALSE TWIN

Creates co-located duplicates on incompatible sheets. The player must target lineage and compatibility rather than appearance.

THE CHATTER ENGINE

Oscillates around a guard boundary. The solution is hysteresis, hold time or a constraint mode - not faster button pressing.

THE ONE-BIT KING

Claims one selector contains geometry, state and identity. The arena is won by splitting those roles into explicit objects.

THE OPTIMIZER

Deletes uncertainty and lineage to make every patch "fast." The final fight uses evidence, conflict traces and reconstruction to prove why a smaller record can be less truthful.

Narrative inference. The characters and themes are original. They dramatize explicit source boundaries: one-bit narrow roles, coordinates not identity, packed hashes not complete lineage, and compression not correctness. [S1 pp. 10, 13, 15](#) [S5 pp. 11-14](#)

MAKE LOGIC INSTANTLY READABLE

Art Direction, Animation and Procedural Audio

Visual language

support

compatibility

guard

commit

rejection

trace

- **World:** clean vector architecture, dark substrate voids, luminous path networks and readable silhouettes.
- **Rule objects:** semi-transparent blocks with physical sockets, phase bands, content seals and animated dependency cables.
- **Preview:** ghost geometry, dotted collision volumes and alternate route overlays.
- **Topology:** sheet separation shown through parallax/layer offset rather than impossible camera tricks alone.
- **Definitions:** resolution-independent SVG/Canvas paths; no texture atlas required for the first slice.
- **Patterns:** Gielis, Lissajous, rose, spiral and periodic-field contours give machines distinct identities without sampled art.

Animation

Rule blocks pulse by state: idle, candidate, verified, rejected and committed. Keyframe clips, easing, ping-pong clocks, crossfades and property-path application are enough for the first release.

Audio grammar

Event	Sound logic
Support admitted	Short low click; one oscillator voice.
Compatibility complete	Second interval joins the first; consonance indicates matched tags.
Guard approaching	Pitch rises with normalized residual, but never pretends to be probability.
Verified proposal	Three-note cadence; still unresolved until commit.
Commit	Full chord plus percussive transient; lineage motif appended to music.
Rejection	Dissonant interval shaped by reason family, followed by a readable text cue.
Replay divergence	Beat sequence splits at the first mismatched hash.

AUDIO TARGET

The player should hear the authority chain.

Procedural oscillator/noise cues and beat sequences keep the package compact and make each verification stage musically legible.

Screen output

The first game can use the 3.9 Canvas renderer, gradients, transforms, z-order, camera, grid, vignette, particles and HUD. More advanced field visualization from the IQ Field profile is optional.

Vector, animation and procedural-audio support: **S6 pp. 5-7, 10, 14** . Pattern families: **S2 pp. 5-7, 16+** .

DEPTH WITHOUT EXCLUSION

Accessibility, Difficulty and Hint Design

Perceptual access

- Color + icon + shape + text for every rule class.
- High-contrast mode and reduced glow.
- Adjustable trace thickness and animation speed.
- Screen-reader labels for definition inspector and reason codes.
- No solution depends only on tiny moving particles.

Motor access

- Snap-to-socket and one-button cable routing.
- Full pause during rule editing in standard mode.
- Hold/toggle options for Rule Lens.
- Remappable named actions across keyboard, gamepad and touch.
- Undo to checkpoint without timing precision.

Cognitive access

- Plain-language layer before formal type detail.
- One new distinction per early room.
- Trace highlights the first divergence, not every intermediate value.
- Reason codes grouped into reference, type, compatibility, guard, conflict and invariant families.
- Glossary terms unlock in context.

Three difficulty profiles

Profile	Editing time	Inspector	Failure assistance	Best for
Guided	World pauses fully.	Plain language, relevant sockets only.	Highlights the first failed gate and suggests one next action.	Story players and accessibility.
Standard	World pauses at benches; hazards continue in late chapters.	Plain + trace layers.	Full reason code and before/after comparison.	Intended first playthrough.
Expert / Audit	Optional live edits and tighter budgets.	Full record, hashes and dependency graph.	No suggested action; deterministic trace remains available.	Optimization, speedruns and workshop validation.

Hint ladder

1. Symptom

2. Relevant phase

3. First failed gate

4. Missing rule class

5. Candidate wiring

6. Full solution

Design goal. The formal substrate should increase explainability, not create jargon walls. The expert layer is optional; the physical world and plain-language rejection should carry the main experience.

ADDITIVE SCHEMA, INSPECTABLE RECORDS

Technical Game Data Model

Project-level additions

The first implementation can extend the 3.9 GameProject additively rather than inventing a separate engine:

```
{
  "schema": "ugts-kc-definition-breaker-0.1",
  "rule_library": [...content-addressed definitions...],
  "rule_instances": [...physical blocks and fixed
sockets...],
  "edit_zones": [...what the player may manipulate...],
  "commit_policies": [...accepted statuses, margins,
conflicts...],
  "puzzle_goals": [...state, route, lineage and replay
predicates...],
  "scenes": [...ordinary 3.9 entities, tilemaps, art and
audio...]
}
```

Canonical rule definition

```
{
  "id": "rule:door-open-on-power-v1",
  "kind": "guarded_transition",
  "domain": "door_state x power_state",
  "codomain": "door_state",
  "evaluation_phase": 7,
  "dependencies": ["guard:power-positive-v1"],
  "parameters": {"threshold": 1.0},
  "invariants": ["door_node_id_stable"],
  "provenance": {"class": "game-authored"},
  "content_hash": "sha256:..."
}
```

This schema is an application-level proposal derived from the 3.6 definition record, 3.0 scene/event records and 3.9 project model.

[S4 pp. 3-7, 11-12](#)

[S3 pp. 6-7, 19](#)

[S6 pp. 3-4](#)

Runtime records

RuleInstance

id, definition_ref, transform, socket_state, ownership, edit_zone, lineage_ref

RuleSocket

domain, codomain, phase, multiplicity, allowed_kinds, lock_state

RuleEditProposal

id, tick, editor, target, operation, definition_ids, expected_pre_hash, confidence, numeric_error, margin, lineage_label

RulePatch

add/remove/replace/reorder/connect/disconnect/parameter/topology/constraint operation

PuzzleGoal

required state predicates, route class, preserved invariants, forbidden reason codes, replay hash policy

RuleTrace

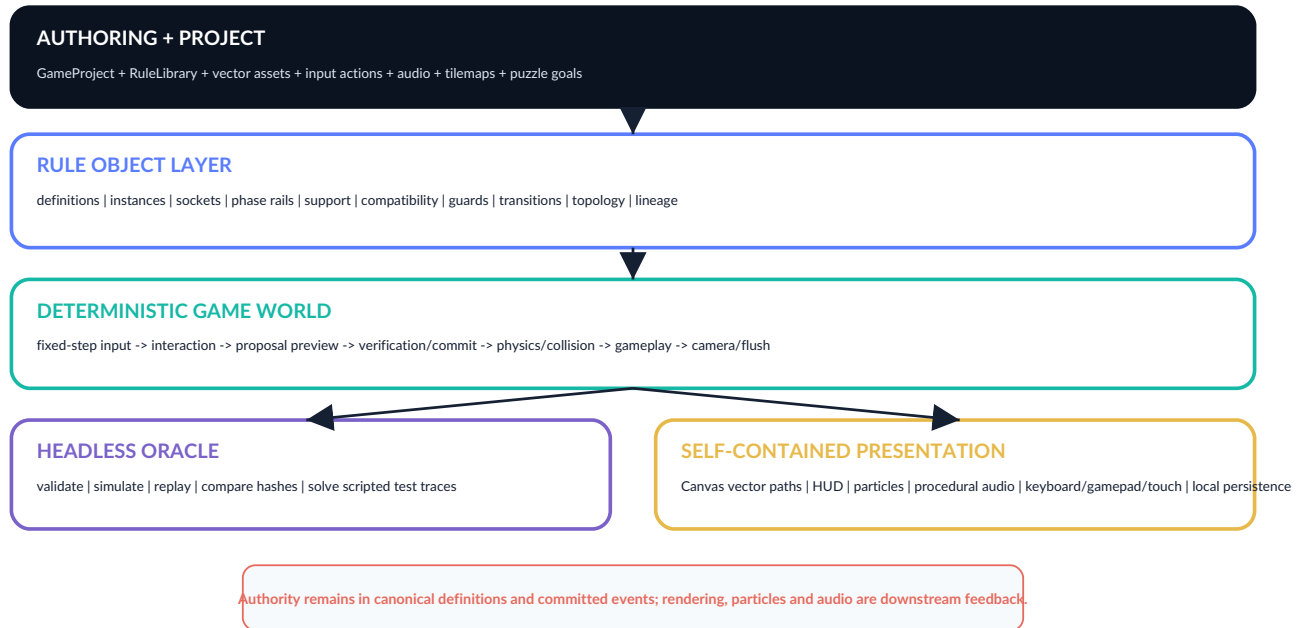
ordered phase outputs, first divergence, gate results, conflict decisions, pre/post hashes

Identity rules

- Definition ID != physical rule-block instance ID.
- Scene-node ID != asset ID != current coordinate.
- Content hash verifies canonical semantics but does not replace lineage.
- Replacing a referenced definition may update many instances; moving one block does not create a new definition.
- Rendered icon and color are downstream views of the record.

ONE PROJECT, TWO EXECUTION TARGETS

Runtime Architecture and System Order



Suggested components

Transform2D, Body2D, Collider2D, RuleCarrier, DefinitionRef, RuleSocket, SupportGate, CompatibilityTags, GuardRelation, TransitionPatch, TopologyState, ConstraintMode, LineageRef, PuzzleGoal, InspectorHint.

Suggested fixed-step insertion

1. Resolve InputFrame.
2. Player controller and rule-object manipulation.
3. Assemble candidate graph and preview requests.
4. Verify/commit accepted rule proposals at a stable pre-physics phase.
5. Physics, bounds and collision.
6. Gameplay hazards, pickups and puzzle-goal checks.
7. Animation, lifetime and procedural feedback.
8. Camera, deferred destruction, snapshot/hash/save.

Implementation decision. Keep the rule system as custom, stably ordered systems inside the existing fixed-step world. Do not let the browser UI mutate entity dictionaries directly. All committed edits go through the same proposal and patch path used by headless tests.

SMALL, DETERMINISTIC AND HONEST

Persistence, Seeds, Compression and Performance

Save/reconstruction strategy

1. Store immutable/vector definitions once and reference them by stable ID/content hash.
2. Generate baseline room geometry from project seed and bounded grammar where reconstruction is deterministic.
3. Store current compact hot state: stable IDs, transforms, modes, phase/sheet/orientation, active constraints and lineage references.
4. Append authoritative player edits, topology changes, conflict decisions and external inputs as events.
5. Create checkpoints when replay cost or divergence risk exceeds the room budget.
6. Verify a loaded save by replaying from the nearest checkpoint and comparing the canonical state hash.

64-bit key use

Use a finite key as a local room/phase index or deterministic challenge code. It may accelerate lookup and prefix refinement, but must not stand in for stable identity, definitions, uncertainty, route state or full lineage.

Compactness advantages

- Vector paths and gradients replace texture atlases for the first release.
- Procedural oscillator/noise cues replace sampled effect files.
- Shared definition records prevent repeated rule payloads.
- Seeded room variation stores parameters and branch choices rather than materialized copies.
- Verified-event compaction avoids dense logging of unchanged candidates.
- Cold definitions/editor metadata stay outside the hot simulation record.

Targets to measure, not claims

Target	Initial design goal
Fixed-step rate	60 ticks/s target on ordinary desktop browsers; measure p50/p95/p99.
Room rule graph	Usually <= 64 active rule instances and <= 256 definition dependencies.
Grammar	Explicit depth/symbol/stack budgets; reject overflow before simulation.
Preview latency	Immediate for local edits; degrade by reducing horizon, not by hiding status.
Web package	Small enough for instant offline opening; exact byte target set after the first slice.

Precision rule. Packed/fixed-point or reduced records are accepted only when coordinate and guard error remain below the declared event margin and do not change ordering. Memory reduction is not correctness. [S1 pp. 9-15](#) [S5 pp. 8-14](#)

SHIP THE SIMPLEST DISTINCTIVE VERSION FIRST

Platform, Modes and Expansion Path

FIRST: OFFLINE HTML5

Desktop browser and static-file distribution using 3.9's Canvas/ Web Audio runtime.

- Keyboard, gamepad, pointer and touch.
- Headless Python validation and simulation.
- Single-file or split bundle.
- Local saves and best-solution records.

RECOMMENDED MVP

SECOND: NATIVE DESKTOP / ANDROID

Port the stable record and commit core only after the puzzle schema is proven.

- Native input and storage.
- Optional field/GPU presentation.
- Phone-scale touch Rule Bench.
- Reuse compact vector/seed assets.

AFTER VERTICAL SLICE

LATER: TWO-HAND / CO-OP

Use bimanual transforms and deterministic ownership for spatial rule manipulation.

- Local co-op “dual authority.”
- VR hand joints and handover.
- Server-authoritative workshop sessions.
- Replay and interest management.

NOT MVP

Game modes

Mode	Core promise	Substrate leverage
Campaign	Authored puzzle-adventure with narrative and bosses.	Full rule vocabulary, fixed-step runtime, checkpoints and progression.
Worldseed Rooms	Daily compact challenge code with verified solution replay.	Bounded grammar, 64-bit index, state hash and input/event recording.
Audit Trials	Optimize a room under rule-count, event-count or lineage constraints.	Reason codes, trace metrics, canonical hashes and deterministic conflicts.
Workshop	Author and share bounded puzzle graphs without arbitrary scripts.	Content-addressed definitions, schema validation, source/asset hashes; optional evidence ledger.
Dual Authority	Two players hold different rule classes and must coordinate commits.	Ownership, handover, proposals, same-time conflicts and replay.

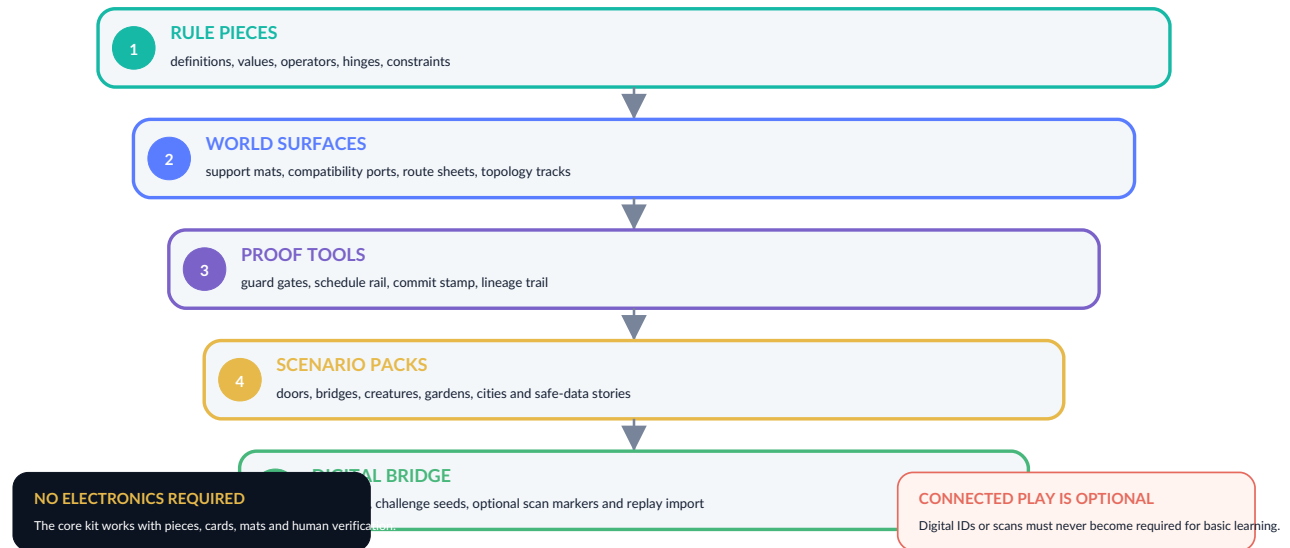
Roadmap boundary. The supplied reports do not claim a finished multiplayer service, native OpenXR binding or general console/ mobile engine. Those remain explicit later engineering tasks. [S3 pp. 15-17](#) [S6 p. 15](#)

A NON-ELECTRONIC FOUNDATION WITH AN OPTIONAL DIGITAL BRIDGE

Definition Breaker Lab: Physical Play Program

DEFINITION BREAKER LAB - PHYSICAL PLAY SYSTEM

A modular toy family in which the same pieces can teach, play, prototype and later map into the digital game.



Purpose

Make formal substrate ideas graspable before screens, coding or symbolic notation. Every activity should work with ordinary human facilitation.

Product strategy

Begin as print-and-play and classroom prototypes. Promote pieces to manufactured toys only after child testing, accessibility review and independent safety engineering.

Digital relationship

Physical and digital products share stable concept IDs and challenge seeds, but neither is a disposable accessory for the other.

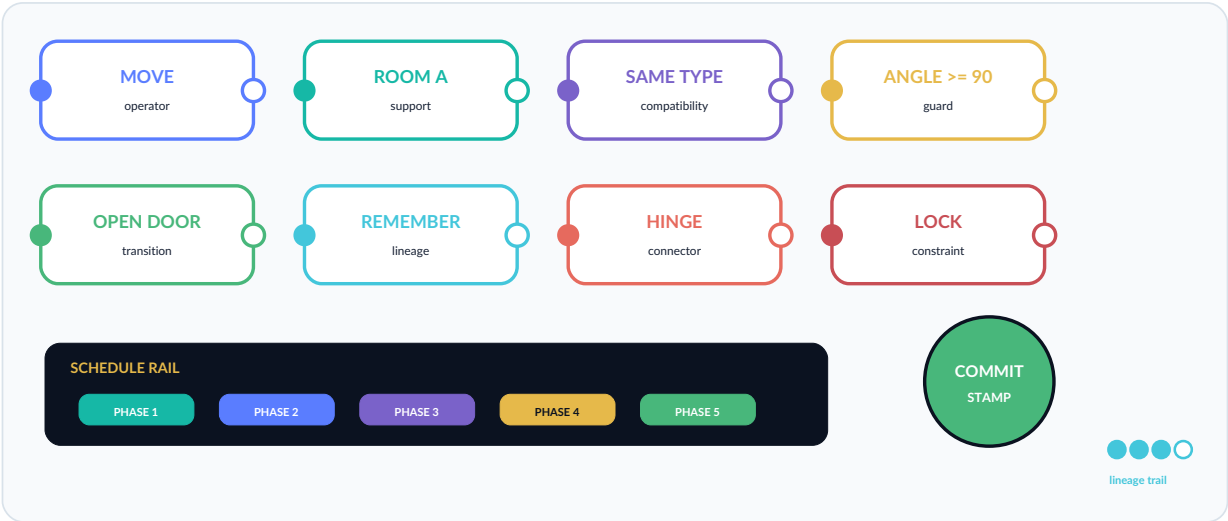
Placeholder status. The following sections are design scaffolding, not manufacturing drawings, certified curricula or finished commercial game rules.

A SHARED PHYSICAL VOCABULARY FOR THE ENTIRE PRODUCT FAMILY

Rule Tile Starter Kit

RULE TILE STARTER KIT - VISUAL PLACEHOLDER

Not manufacturing artwork. The drawing shows roles, scale relationships and interaction grammar.



Proposed starter contents

24 rule tiles	values, operators, supports, compatibility, guards, transitions, hinges and constraints
8 instance pawns	stable IDs distinct from reusable rule/asset cards
4 support mats	room, circle, cone-like wedge and nested shells
1 schedule rail	ordered slots with reversible labels
1 try tray	candidate arrangement clearly separated from committed world
1 commit stamp	green token or card, never an irreversible ink requirement
40 event cards	lineage, rejection reason, before/after and replay sequence

Piece design grammar

- Color + icon + text + edge shape; never color-only coding.
- Large, graspable pieces with rounded edges and high-contrast labels.
- Input sockets physically differ from output tabs.
- Rule ID remains readable even when a child covers the center.
- Back side reveals the deeper Builder/Designer information.
- No hidden electronics in the starter kit.

DIMENSION PLACEHOLDER

Target rule tile size: [__ mm x __ mm x __ mm]
Minimum readable text: [__ pt at __ cm]
Graspability test group: [__]
Material prototype: [card / foam / wood / molded polymer]

THE CANONICAL AUTHORITY SEQUENCE BECOMES TABLE FURNITURE

World Surfaces, Compatibility Ports, Guard Gates and Schedule Rails

Support mats

Define where a rule may act: room, circle, wedge, shell, route strip or custom finite region.

Prototype question: Can a child identify “inside support” without text?

Compatibility ports

Interchangeable badges for sheet, phase, team, role, ownership and mode.

Prototype question: Do learners understand that matching location and matching compatibility are separate?

Guard gate

A line, arch, dial or balance that can show crossing, touch, stay, margin and hysteresis.

Prototype question: Can the facilitator demonstrate noisy chatter without fragile electronics?

Schedule rail

Ordered phase slots for construct, transform, support, compatibility, guard, transition and record.

Prototype question: Are legal-but-noncommuting orders obvious enough to compare?

Try tray

A physically separate area for candidate edits. Pieces here have no authority until verified.

Prototype question: Is preview versus committed world unambiguous from across a table?

Commit and reject tokens

Green commit, reason-coded reject and optional “unknown” token. Rejection is information, not punishment.

Prototype question: Do children voluntarily use the reason code on their next attempt?

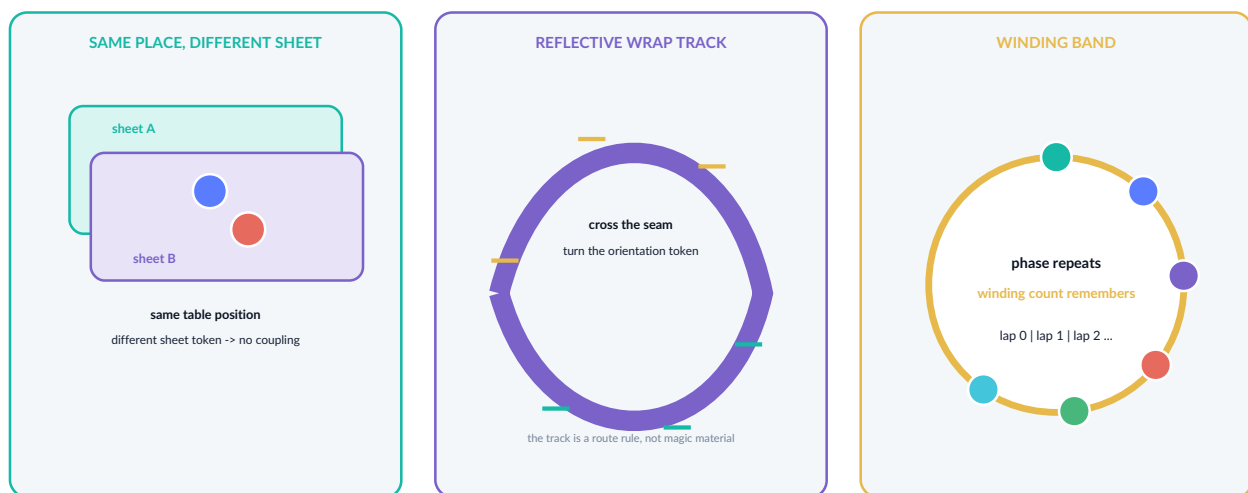
Starter lesson: A movement tile sits on the table but outside the player mat. Ask the support question first. Then move it inside but give it the wrong role badge. Ask compatibility second. Finally place the guard gate and test the crossing.

MAKE ROUTE IDENTITY VISIBLE BEFORE USING THE WORD TOPOLOGY

Topology Tracks, Sheets and Winding Bands

TOPOLOGY TRACKS, SHEETS AND WINDING BANDS

The toys make route history visible instead of treating a crossing on the table as an automatic collision.



Sheet overlays

Transparent or offset boards let pieces occupy the same projected coordinate while staying distinct. A compatibility bridge can temporarily connect sheets.

Wrap tracks

Boundary pieces encode ordinary wrap, half-turn bundle or reflective wrap. Orientation tokens change only when the printed route rule says so.

Winding bands

Beads or clips count laps separately from current phase. Re-entering the same location does not erase route history.

Topology boundary. These are routing and identification toys. They do not claim physical Klein bottles, impossible self-intersection, energy creation or magical materials.

A PHYSICAL BRIDGE FROM RELATION MAINTENANCE TO CONSTRAINT RELEASE

Hinges, Constraints and the Missing Shackle Set

Proposed pieces

- **Pivot hinge:** one fixed point and a rotating arm.
- **Parity hinge:** a guarded two-position selector for route A or B.
- **Connector hinge:** a structural link with no numeric magnitude.
- **Constraint clip:** keeps two pieces in a declared relation over time.
- **Shackle row:** one removable constraint among several.
- **Tangent slider:** allows motion along a boundary but blocks normal motion.
- **Two-hand beam:** marked grips for midpoint, scale, alignment and handover play.

MECHANISM PLACEHOLDER

Connector type: [slot / pin / textile loop / hook-and-loop]

Allowed degrees of freedom: [__]

Release force target: [__]

Pinch-point review: [pending]

Missing Shackle challenge

1. Build a three-direction slider with two constraint rows.
2. Ask learners to predict the legal motion.
3. Remove one marked row.
4. Count the newly available direction.
5. Record the freedom gain on an event card.
6. Explain why one extra degree of freedom does not prove chaos.

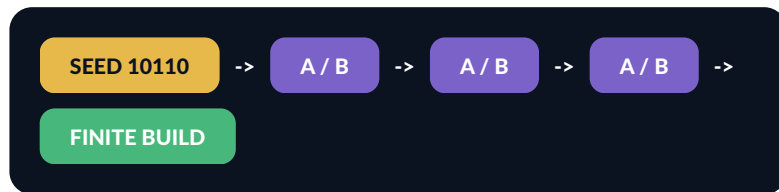
Advanced extension: Place the slider on a curved zero-boundary track. Propose a move, project it onto the tangent direction and compare the intended and legal motion.

Safety boundary. Use low-force, soft-edged prototypes. No spring-loaded release, load-bearing use, body restraint or mechanical trapping is part of this concept.

COMPRESSION BECOMES RECONSTRUCTIBLE PLAY, NOT A MAGIC CLAIM

Seed and Bounded Grammar Construction Kit

Kit structure



- Branch cards choose one of two declared productions.
- Depth tokens limit the number of levels.
- Symbol counters and stack pieces make budgets visible.
- Prefix strips progressively narrow a cell or route family.
- A challenge key selects an address; separate cards still hold definitions, uncertainty and lineage.

Reconstruction race without speed scoring. Two teams receive the same seed and rule deck. They build independently, compare final objects and replay the first different branch if the results diverge.

What children should say

GOOD

"The seed tells us which branch to choose under this finite rulebook."

GOOD

"The key helps us find a cell, but we still need the definitions and history."

AVOID

"The short number contains the whole infinite world."

AVOID

"A binary branch is automatically chaotic."

Physical adaptation of bounded grammar, prefix refinement and finite-key boundaries from [S5 pp. 7-15](#).

THE FIRST MANUFACTURABLE GAME SHOULD TEACH THE AUTHORITY CHAIN

Tabletop Game Scaffold: Rule Repair Crew

RULE REPAIR CREW

Cooperative family puzzle game for 2-5 players

ENTRY Explorer / Builder

TIME 20-35 min

CORE support, compatibility, guard, commit, lineage

Round structure

1. Reveal a broken room and its current committed rules.
2. Each player receives a role: carrier, support checker, compatibility checker, guard keeper, recorder.
3. The team may move only a limited number of rule pieces into the try tray.
4. Predict the result and run the verification sequence.
5. Accept one commit or receive a reason-coded rejection.
6. Spend no penalty for a well-explained rejection; the team simply revises.
7. Win by reaching the room goal within the lineage-card budget.

Scenario ladder

1. Wrong mat	Support only
2. Wrong badge	Add compatibility
3. Almost crossed	Add guard status and margin
4. Order rail	Add noncommuting operations
5. Broken dependency	Add definitions and hinges
6. Same place, two sheets	Add topology and route identity

UNRESOLVED DESIGN

Scoring model: [cooperative mastery / campaign stars / lineage efficiency]

Randomness: [none / bounded scenario draw]

Replay storage: [cards / photo / optional app]

A COMPACT-CODE GAME THAT TEACHES WHAT THE CODE DOES NOT CONTAIN

Tabletop Game Scaffold: Worldseed Builders

WORLDSEED BUILDERS

Reconstruction and procedural-design game for 1-4 players

ENTRY

Builder / Designer

TIME

25-45 min

CORE

seed, bounded grammar, prefix, winding, definitions

Build phase

Read one seed bit at a time, choose the declared branch, spend depth/symbol tokens and place the resulting route or rule object.

Audit phase

Swap builds with another player. Verify definitions, branch history, budgets, winding and final address.

Repair phase

If the builds differ, replay prefixes until the first divergent choice or missing definition appears.

Modes

Mode	Goal	Learning evidence
Mirror Build	Two players reconstruct the same world independently.	Matching result plus identical branch trail.
Broken Archive	Recover one missing definition that prevents reconstruction.	Explanation of why the seed alone was insufficient.
Smallest Proof	Use the fewest event cards while preserving replay.	Reconstruction succeeds from a checkpoint plus novelty events.
Designer Mode	Author a fair finite grammar and publish a challenge code.	Declared budgets, kill criteria and solvability test.

Commercial boundary. Do not market the seed as infinite content or one-number world ownership. The core delight is transparent reconstruction.

COOPERATION BECOMES AN AUTHORITY PROBLEM, NOT JUST A COMMUNICATION PUZZLE

Tabletop Game Scaffolds: Dual Authority and Replay Detective

DUAL AUTHORITY

Two players hold different rule classes and neither can commit alone.

- Player A controls support and movement.
- Player B controls compatibility and guards.
- Both may propose in parallel.
- A shared verifier orders conflicts.
- A handover token changes ownership without changing object identity.

OPEN QUESTION

Should communication be unrestricted, icon-only or turn-limited?

REPLAY DETECTIVE

One team builds and records a solution. Another replays from a checkpoint.

- Every event has sequence, source and before/after state.
- One optional "corruption" card changes a step.
- Detectives stop at the first mismatch.
- Points reward precise divergence explanation, not speed.
- Advanced mode adds simultaneous events and conflict policy.

OPEN QUESTION

Use physical hashes as short color-shape seals or simple code words?

Design benefit. These games teach that shared systems need explicit authority, ownership, ordering and correction - concepts often hidden in ordinary board games.

A BLANK CONTRACT FOR FUTURE TOYS, GAMES AND CLASSROOM ACTIVITIES

Reusable Prototype Scaffold

PROTOTYPE NAME	[_____]
SOURCE CONCEPT	[version / mechanism / page / design inference]
LEARNING QUESTION	[What careful question should the learner be able to ask?]
ENTRY TRACK	[Explorer / Builder / Designer / Mentor]
PHYSICAL PIECES	[list, count, proposed material, large/small-part flag]
SETUP	[starting world, committed rules, player roles]
LEGAL PROPOSALS	[connect / reorder / replace / move / wrap / release]
VERIFICATION	[support, compatibility, guard, conflict, invariant, commit]
REJECTION REASONS	[specific reason codes a learner can act on]
LEARNING EVIDENCE	[prediction, explanation, replay, authored challenge]
DIGITAL MAPPING	[stable IDs, record types, optional challenge seed]
ACCESSIBILITY	[tactile, visual, language, motor and cooperative alternatives]
SAFETY / PRIVACY	[hazards, supervision, data excluded, standards review]
KILL CRITERION	[When is an ordinary toy or lesson clearer than this substrate mapping?]
STATUS	[idea / paper / foam / pilot / revised / independent review]

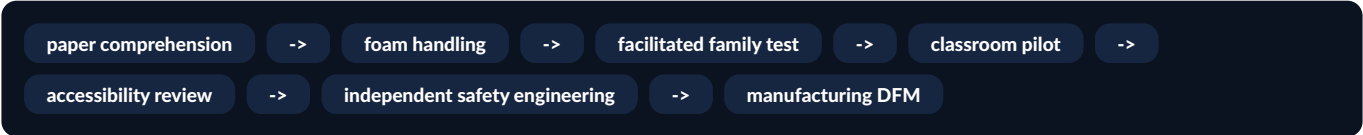
Use this as a gate. A toy concept does not enter the product line merely because it can be decorated with UGTS terms. It must create clearer learning or more interesting play than a conventional alternative.

BUILD EVIDENCE BEFORE COMMITTING TO MATERIALS OR TOOLING

Placeholder Bill of Materials and Validation Plan

Subsystem	Placeholder components	Material / process candidates	Validation still required
Rule tiles	[24-48] large tiles, reversible labels, typed tabs	laminated card, EVA foam, wood, molded polymer	graspability, edge durability, label abrasion, washability, reading distance
World surfaces	folding mats, overlays, shells, track strips	textile, coated paper, silicone-like sheet, rigid board	slip resistance, fold memory, contrast, tactile boundaries
Connectors	slots, clips, textile loops, hook-and-loop	avoid magnets in early prototypes unless separately reviewed	pinch points, pull force, ingestion risk, wear fragments
Mechanisms	hinges, constraints, shackle rows, sliders	low-force foam/card pivots before rigid mechanisms	trapping, sharp edges, breakaway, misuse and body-restraint scenarios
Proof tools	try tray, guard gate, commit/reject/unknown tokens	cardboard and large tokens	authority distinction visible, reason codes understandable
Replay	event deck, sequence clips, checkpoint cards	paper/card, optional dry-erase surface	record completeness, replay success, privacy-safe use
Storage	sorted tray, labels, educator inventory	box insert or pouches	setup time, missing-piece detection, classroom reset
Optional digital bridge	printed IDs, QR-like marker or manual code	print only at first; NFC/electronics later if justified	privacy, offline operation, scan failure, no hidden auto-commit

Prototype validation sequence



PHYSICAL PLAY ADDS OBLIGATIONS THAT THE DIGITAL CONCEPT DOCUMENT DID NOT HAVE

Safety, Inclusion and Manufacturing Boundary

Child safety

- Prototype with large soft pieces first.
- Age-gate small parts and complex mechanisms.
- Review pinch, trap, sharp-edge, cord, magnet, battery and breakage risks.
- No load-bearing, body-restraint or projectile play.
- Final products require target-market standards review and independent testing.

Inclusive access

- Color + shape + icon + text coding.
- Tactile and high-contrast alternatives.
- Read-aloud and low-literacy facilitation.
- Seated tabletop reach and reduced-force handling.
- Cooperative scoring, no required speed.
- Quiet mode and simplified visual field.

Data and ethics

- Offline by default.
- No child account required for core play.
- No personal identifiers in challenge codes.
- No wallet words, passwords or financial secrets.
- Photos and portfolios stay under educator/family policy.
- Unknown and opt-out states are valid.

Manufacturing non-claim. This document does not certify compliance with any toy, educational, accessibility, privacy or product-safety standard. It supplies design questions and prototype gates only.

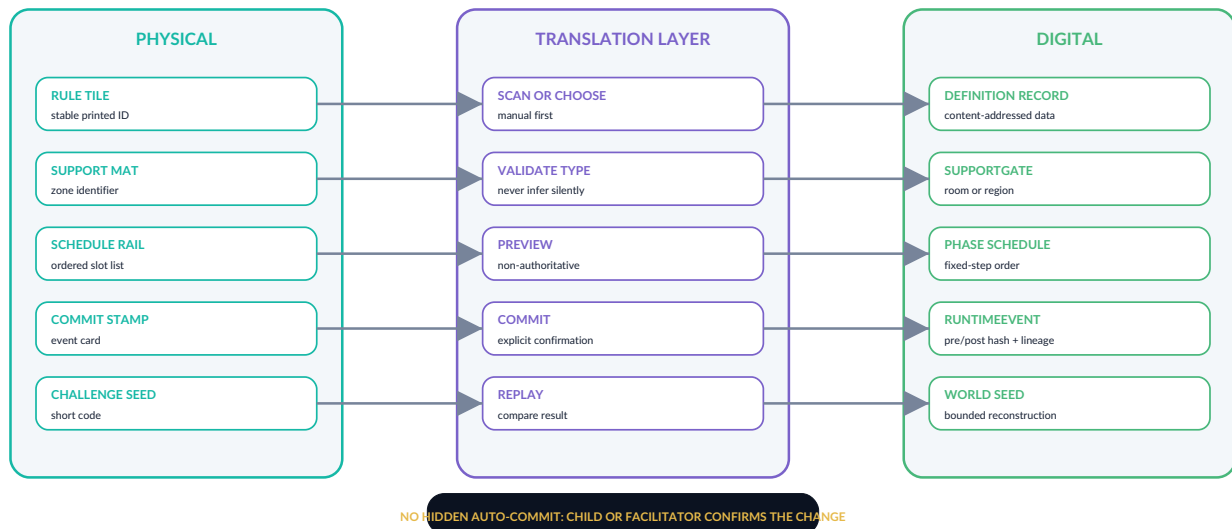
Inclusion acceptance criterion. A learner must be able to understand support, compatibility, guard and commit without seeing color, reading fine print, moving quickly or using a connected device.

CONNECTED PLAY SHOULD MAKE TRANSLATION VISIBLE, NOT MAGICAL

Toy-to-Digital Bridge and Future Connected Play

FROM TABLETOP PIECES TO THE DIGITAL WORLD

The bridge is optional and reversible: every physical concept remains understandable without a device.



Recommended first bridge

1. Each physical tile has a human-readable stable ID.
2. The learner manually chooses the same tile in the digital Rule Bench.
3. The game validates the type and shows a ghost preview.
4. The learner confirms commit.
5. The digital replay can print or display a physical event-card sequence.

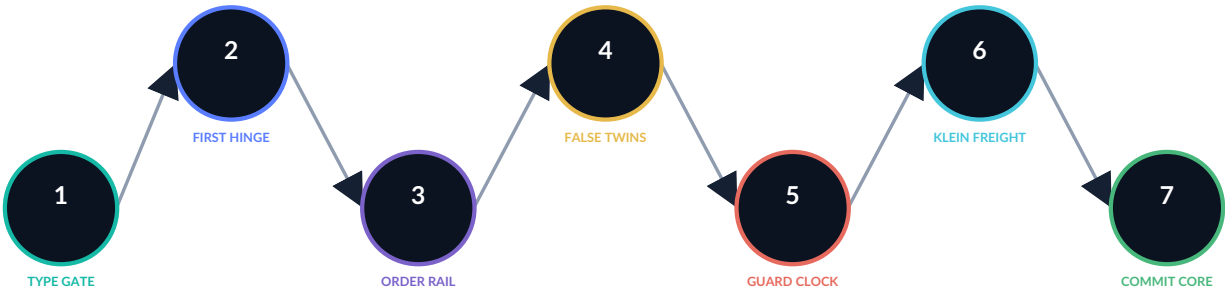
Later options, only if useful

- Camera recognition of large printed markers.
- Challenge-code import from physical Worldseed cards.
- Optional NFC for durable classroom inventories.
- Projection of digital support zones onto a physical mat.
- Two-device cooperative authority.

Never hidden: recognition confidence, unresolved pieces, candidate versus committed state, and every imported rule ID.

THE SMALLEST RELEASE THAT PROVES THE IDEA

Recommended Vertical Slice



VERTICAL SLICE TARGET

20-30 minutes | 7 rooms | 1 rule bench | 1 deterministic replay reveal | 1 topology boss | offline HTML5 build

Playable scope

- One compact facility wing, 7 rooms, 20-30 minutes.
- Player movement, dash, health/hazard fallback and camera from 3.9.
- Rule Lens with simple and trace layers.
- Commit Bench supporting connect, disconnect, reorder and replace.
- Rule objects: literal, operator, hinge, compatibility seal, guard and transition.
- One deterministic replay sequence and one same-tick conflict.
- One orientation-wrap boss using sheet/orientation state.
- Self-contained offline HTML5 build plus headless smoke trace.

Acceptance metrics

Gate	Pass condition
Clarity	New players solve the first three rooms without opening the formal record layer.
Authority	No UI or presentation path mutates canonical state outside verified commit.
Replay	Every accepted room solution reproduces its final state hash from checkpoint.
Reason codes	Every rejected edit reports one primary cause and exact target/reference path.
Boundedness	No room can create unbounded grammar, event chatter or dependency growth.
Delivery	Project validates, simulates headlessly and opens offline in a browser.
Fun	At least one room produces an "I changed the law" moment rather than a cable-matching chore.

VERTICAL-SLICE DECISION

Prove Definition -> Physical Object -> Preview -> Verified Commit -> Changed Traversal.

Do not begin with a general-purpose visual programming editor. Build seven authored rooms around a narrow rule taxonomy, then expand only after the loop is enjoyable.

BUILD, MEASURE, AND STOP WHEN THE PREMISE STOPS HELPING

Implementation Phases, Kill Criteria and Source Register

Implementation phases

0. Contract	Rule schema, canonicalization, definition hashes, dependency validation, proposal and patch records.
1. Tactile loop	Physical rule blocks, sockets, Rule Lens, preview state and verified commit in one test room.
2. Puzzle proof	Four rooms: type, order, compatibility and guard. Headless replay and reason-code tests.
3. Distinctive content	Topology room, conflict/replay room, boss, procedural audio and final art pass.
4. Productization	Campaign shell, accessibility, saves, challenge codes, build reports, manifests and offline packaging.

Kill or simplify when...

- Players must read raw JSON to understand normal rooms.
- The editable graph grows faster than support/compatibility can prune it.
- Invalid edits return generic failure rather than a useful reason.
- Visual preview cannot clearly distinguish candidate from committed state.
- Packed precision or cross-platform behavior changes event order.
- Topology is only cosmetic and never alters route, orientation or compatibility.
- Most content is cheaper and clearer as ordinary authored scripting.
- The four-room prototype feels like wiring homework instead of world transformation.

Source register

ID	Source and most-used pages
S1	UGTS-GN 1.1, pp. 2-5, 9-15, 18-19, 28: canonical authority, identity, packed records, precision, memory, one-bit boundaries.
S2	UGTS-KC 2.0, pp. 4-15, 21: pattern/field roles, route identity, hybrid events, degeneracy, memory and final definition.
S3	KC Two Hands 3.0, pp. 5-13, 15-19: scene identity, physical interaction, proposal/commit, replay, records and roadmap boundaries.
S4	UGTS-KC 3.6, pp. 1-14: content-addressed definitions, dependency graph, hinges, operation order, traces and kill criteria.
S5	UGTS-KC 3.6.2 SCLP, pp. 5-15: interval guards, winding, reflective wrapping, constraint release, bounded grammar and 64-bit layouts.
S6	UGTS-KC 3.9, pp. 3-15: vector art, input, animation, tilemaps, audio, fixed-step game world, persistence and HTML5 export.
S7	UGTS-KC 3.9.3 IQ Field, pp. 16, 23: optional richer field primitives and bounded procedural/query status.
S8	KC Atlas 3.9.3 / Spatial Evidence 4.0, pp. 4-7, 11, 17-18: optional evidence, revision, unknown-state and publication ledger for workshop content.
S9	UGTS-KC 3.6.3 SARA, pp. 1-12: optional older-learner public/secret typing, checksum boundary, authorization, public-only certificates and explicit prohibited scopes.
S10	Definition Breaker DB-0.1: immediate concept baseline retained and expanded by this DB-0.2 all-ages learning and physical-play edition.

Final evidence boundary. This design preserves source terminology and documented limits, but all game loops, rooms, characters, bosses, UI, progression, age tracks, lesson plans, toy pieces, tabletop rules, targets and implementation sequencing are original application inferences. No novelty, patentability, commercial viability, completed-game, certified-curriculum or safety-certified-toy claim is made by this PDF.

Definition Breaker is not a game where anything is possible. Every strange change must be explained before it can become real.